ICT378 Cyber Forensics and Incident Response
Final Assignment

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Introduction

Two travellers, John Fredricksen and Jane Esteban, were stopped at Wellington International Airport. The Australian Customs and Border Protection Service launched a thorough digital forensic investigation. Customs agents found one kilogram of methamphetamine hidden inside the lining of a suspect's suitcase, confirming Australian government intelligence suggesting possible involvement in international drug trafficking. Jane's remarks and information about a third person, Steve Kowhai, led detectives to a Petone location where more drugs, firearms, and a desktop computer were found.

The drive images and volatile memory dumps from two laptops and one desktop computer of all three suspects' devices were submitted for forensic analysis in order to assess the degree of the suspects' involvement and establish their motivations.

The examiner conducted the investigation with a range of forensic tools, including Autopsy 4.22.1, Magnet AXIOM, and Windows CMD hashing applications for evidence integrity verification. To ensure evidence integrity, each image drive was processed in a controlled environment, hashed before analysis, and cross-checked during the examination. While Magnet AXIOM was used for memory analysis, communication recovery, and cross-data source correlation, Autopsy was primarily utilised for file system analysis, lost file recovery, web history analysis, and system artefacts.

During the investigation, several operational issues were experienced. Large photographs had to be re-ingested due to inaccurate data source selection in Autopsy, and the workflow was slowed by frequent processing crashes and partially corrupted evidence segments. Due to Autopsy's inherent limitations when parsing volatile memory, memory dump analysis has to be offloaded to Magnet AXIOM. Despite these obstacles, the examiner was able to recover a number of relevant artefacts, including Discord conversations, browser histories, steganographic content, suspicious documents, anti-forensics-related activity, spyware traces, individual browsing activity, and partially erased content.

This report outlines the comprehensive forensic process, encompassing the investigation of all six evidence sources and the findings regarding the potential criminal collaboration, behavior, communications, and concealment methods of the three suspects. To ensure forensic soundness, repeatability, and legal admissibility, the investigation's goals are to recreate user behaviour, locate evidence of drug-related operations, and determine each suspect's roles and affiliations solely through digital evidence.

Executive Summary

This report presents the results of a digital forensic analysis of three devices confiscated during the 2019 Narcos investigation. The New Zealand Customs and Border Protection Service collaborated with the Australian government and launched the investigation after John Fredricksen and Jane Esteban arrived in Wellington, New Zealand, with one kilogram of methamphetamine hidden in their luggage. Additional drugs, guns, and a desktop computer were discovered during on-site inspections, and additional intelligence linked a third person, Steve Kowhai, to the operation.

The drive images and volatile memory dumps of Fredricksen, Esteban, and Kowhai were among the six digital evidence sources made available for analysis. Every piece of evidence was examined using forensic procedures, including initial hashing with Windows CMD tools and processing with Autopsy 4.22.1 and Magnet AXIOM. These technologies enabled file system examination, lost content recovery, user activity reconstruction, and volatile memory analysis.

A variety of associated assets were discovered during the investigation. Discord discussions, browser activity, steganographic content, suspicious documents, prominent personal site traffic patterns, and evidence of antivirus scans, spyware traces, and deleted files were all discovered while studying drive images and memory dumps. Together, these objects give insight into behavioural indicators, communication patterns, operational planning, and attempts to conceal or obfuscate evidence. John's system was linked to indications of steganographic techniques, which is consistent with intelligence reports that he attempted to conceal operational documents within image files. Discord evidence between John and Steve on Jane's device revealed communications and discussions of the logistics of the methamphetamine shipment and its destination.

Despite several challenges, including ingestion failures, tool crashes, damaged image segments, and Autopsy's limitations in parsing volatile memory, the examiner was able to reconstruct a timeline of relevant events and identify linkages between the three suspects. The data revealed on the devices supports coordinated communication, digital concealment, and activities associated with illegal narcotics trafficking.

This report serves to provide an overview of the inquiry as well as the types of evidence found. The following sections of this study contain additional information on the methodology, artefact recovery methods, and complete findings.

List of Acronyms and Tools

Acronyms	Definition
AFP	Australian Federal Police
APT	Advanced Persistent Threat
CMD	Command Prompt (Windows Command-Line Interface)
DFIR	Digital Forensics and Incident Response
FTK	Forensic Toolkit
MD5	Message Digest Algorithm 5 (Hashing Function)
SHA-256	Secure Hash Algorithm 256-bit
OS	Operating System
VM	Virtual Machine
JSON	JavaScript Object Notation (Memory Dump Artefact Format)
LNK	Windows Shortcut File
AV	AntiVirus

Tools	Version	Purpose	Use in Case
Autopsy	4.22.1	File System Analysis, Recover Deleted Files, Obtain Brower History	Used to search disk images for user activity, browser data, filesystem artefacts, suspicious documents, deleted content, and directory structures.
Magnet Axiom		Memory Dump Analysis	Used to analyse volatile memory dumps, recover Discord conversations, recover system-level artefacts, detect spyware traces, and correlate cross-artifact data that Autopsy does not comprehend.
Windows Command Prompt	Built-in	Hashing Commands, Directory Validation, Manual Verification	Used to execute [certutil -hashfile] to perform MD5/SHA hashing and verify the integrity of evidence sources.
SQLite	3.13.1.0	Database Management	Used to edit and ensure the image files are linked to the appropriate cases
FTKImager			

Methodology Details

The investigation followed an organised digital forensic process based on the notions of forensic soundness, integrity preservation, and repeatability. Following evidence collection, the examiner used the Windows Command Prompt (certutil) to generate MD5 hash values to ensure the integrity of all disc images and memory dumps. Each given image consisting of Narcos-1.001, Narcos-2.001, Narcos-3.001, and all memory dumps were hashed separately, and the results were compared to the values specified in the assignment brief. The analysis continued under the assumption that the supplied files contained the complete and accurate dataset for the case, despite the fact that none of the submitted photographs matched their expected MD5 values. To ensure consistency between hashing utilities, these hashes were previewed and checked with FTK Imager.

Following verification, each memory dump and device image was imported separately into Autopsy 4.22.1 for detailed analysis. Autopsy was used to study file system structures, extract deleted files and recycle bin detritus, recover online browsing history, rebuild Discord LevelDB databases, identify orphaned files, analyse photos and EXIF metadata, and do keyword searches. The utility also allowed you to examine registry hives and other application traces like CCleaner, antivirus logs, and steganography tools. A list of words linked to drug distribution, communication protocols, encryption, steganography, and the names of the three suspects were utilised to conduct keyword searches over the entire dataset. This ensured comprehensive coverage across all file locations, including orphaned and unallocated space.

Magnet AXIOM was utilised as a secondary analysis tool because Autopsy's capacity to process volatile memory was limited. Memory dump parsing, volatile artefact extraction, and the reconstruction of Discord chat fragments that Autopsy was unable to fully recover were all made possible by AXIOM. Additionally, artefacts including steganographic residues, image fragments, and in-memory keyword hits were found using AXIOM's sophisticated search and carving capabilities. To guarantee analytical consistency, the results from both tools were cross-validated.

Management of the Autopsy case database required additional intervention. Because Autopsy saves evidence paths as absolute references, shifting the case between systems caused broken linkages. SQLite Database Browser was used to inspect and manually edit Autopsy's internal database entries, relinking disc pictures to their ingestion records. These modifications were exclusively limited to course correction and did not change any evidence material, guaranteeing forensic soundness throughout the procedure.

Several constraints have an impact on the workflow. During the absorption of the larger disc images, Autopsy crashed several times, necessitating incomplete processing. Initial data-source selection mistakes resulted in evidence that was incorrectly connected and had to be manually repaired using SQLite. Furthermore, Autopsy was unable to efficiently read volatile memory

files, thus they had to be re-analyzed using AXIOM. Some artefacts, such as Quasar keylogging files and CCleaner logs, were only discovered in $OrphanFiles, indicating that they had been wiped or destroyed and so lacked entire contextual metadata.

Regardless of the difficulties, the inquiry adhered to formal forensic criteria. Instead of changing the original evidence, all analysis was performed on mounted images or Autopsy-managed copies, hashes were verified before investigation, and conclusions were evaluated using a number of approaches to ensure dependability.

Finding 1 - Analysis of Steve's Device (Narcos-1.001)

Windows artefacts for Narcos-1 (SK-DESKTOP, owner "Steve" per Operating System Information) were used to corroborate Steve's involvement with encryption tools, system cleaning, steganography and the file secret. Autopsy's Windows artefact modules (Run Programs, Web Downloads, Web History, Recent Documents, USB Device Attached and OS Accounts) were correlated to build a consolidated timeline.

OS Accounts data shows that a domain user account steve was created on 29 January 2019 at 03:35:06 SGT, providing attribution for subsequent user-level activity. Shortly after, Web History reveals that Microsoft Edge accessed file:///C:/Users/Steve/Documents/secret at 29 January 2019, 23:44:31 SGT, while Recent Documents records a shortcut secret.lnk pointing to C:\Users\Steve\Documents\secret. Examination of the file system entry for this path now shows only an unallocated, zero-byte stub with all NTFS timestamps cleared (0000-00-00 00:00:00) and hashes matching an empty file (MD5 d41d8cd98f00b204e9800998ecf8427e, SHA-256 e3b0c4…7852b855). This combination of artefacts confirms that a file named secret previously existed and was opened by Steve, but its contents have since been removed beyond recovery.

Windows Web Downloads and Run Programs outputs further corroborate Steve's acquisition and execution of tools consistent with concealment and clean-up activity. On 30 January 2019, Web Downloads shows TrueCrypt Setup 7.1a.exe retrieved from softpedia-secure-download.com at 07:27:43 SGT and ccsetup552.exe (CCleaner installer) from download.ccleaner.com at 07:40:51 SGT. At 08:04:00 SGT the same morning, Run Programs records concurrent execution of \Program Files\TrueCrypt\TrueCrypt.exe, \program files\ccleaner\ccupdate.exe, \users\steve\downloads\ccsetup552.exe and \program files\ccleaner\ccleaner64.exe, all under the Steve user, with significant network usage. This shows that Steve not only downloaded but actively used both encryption and wiping/clean-up utilities.

On 1 February 2019, Web Downloads records C:\Users\Steve\Downloads\Image Steganography 1.3.0 setup.exe being obtained from sourceforge.net at 08:16:32 SGT. Web History shows multiple hits to ProtonMail around the same time (searches for ProtonMail followed by visits to https://protonmail.com/, https://mail.protonmail.com/login and …/inbox between 08:12:23 and 08:22:06 SGT). Run Programs then records execution of \Users\Steve\AppData\Roaming\CPascoe\Image Steganography\ProgData\Image Steganography.exe at 08:27:00 SGT. Taken together, these artefacts corroborate that Steve acquired and ran a steganography application while actively logged into a ProtonMail inbox, consistent with preparing or processing concealed content.

Finally, USB Device Attached artefacts on Narcos-1 show several external storage devices being connected during the relevant period, including a Toshiba USB device at 29 January 2019

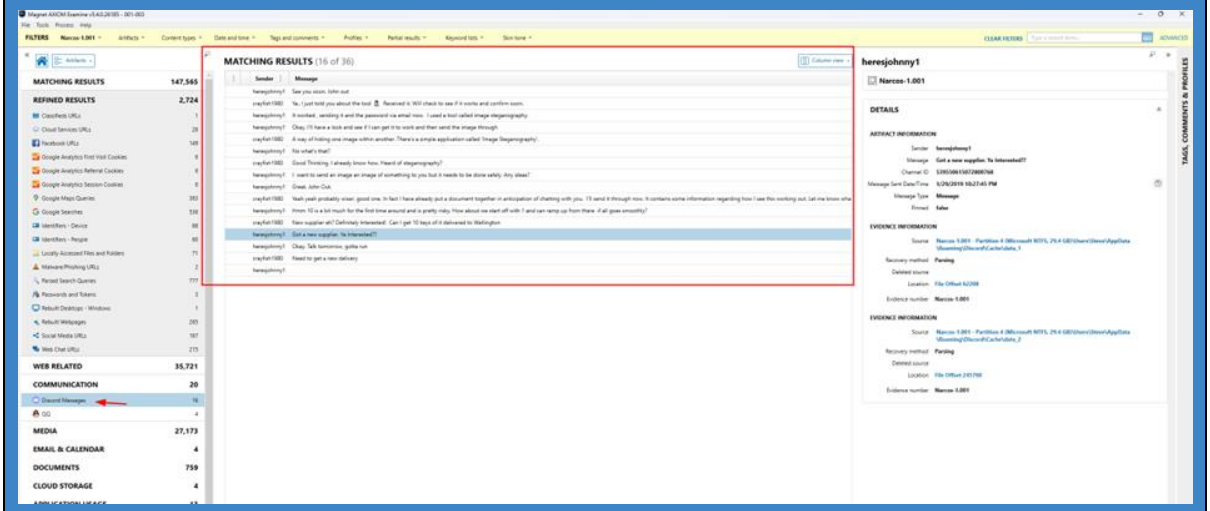
12:01:20 SGT, a Seagate Backup device at 31 January 2019 11:04:08 SGT and Western Digital portable drives at 1 February 2019 10:41:46 SGT and 2 February 2019 10:56:41 SGT. These attachments overlap temporally with Steve's use of TrueCrypt, CCleaner, and Image Steganography, providing a plausible channel for moving or exfiltrating protected or concealed data. Overall, the Windows artefacts independently reinforce the conclusion that Steve accessed a now-wiped file called secret, employed encryption and cleaning tools, used steganography software, and had access to external storage during the critical time window.

Finding 1.1 Communications (Discord Logs)

The Discord log file 000006.log was found on Steve's device in the directory: "/img_Narcos-1.001/vol7/Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/".
The recovered log comprises communication between two persons about logistical plans for a narcotics sale. The content includes information about medicine quantities, distribution locations, and operational strategy. The communications also includes an explicit discussion about the usage of steganography to disguise sensitive information.

Relevant messages include:

- Drug Quantity and Delivery Location:
  "New supplier eh? Definitely Interested! Can I get 10 keys of it delivered to Wellington"
  - Evidentiary Interpretation: Suggests coordination for the smuggling of methamphetamine.
- Encryption Method:
  "A way of hiding one image within another. There's a simple application called 'Image Steganography"
  - Evidentiary Interpretation: Indicates awareness and use of digital concealment techniques.
- Alternative Location:
  "Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone."
  - Evidentiary Interpretation: Correlates with the intelligence obtained during the interrogation and initial investigation



File Name	000006.log
Analysis	Discord messages
File Path	/img_Narcos-1.001/vol7/Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/
File Size	34411
Created Time	2019-01-29 11:43:06 SGT
Accessed Time	2019-02-02 10:38:47 SGT
Modified Time	2019-02-02 10:38:47 SGT
MD5 Hash	19da7fc1b8da010f787b4671aad2570f

The Discord logs contain direct communication evidence that links Steve to the planning and coordination of drug-related actions. The evidence of steganography, delivery locations, and narcotics quantities corroborates additional digital artefacts and contribute to the larger narrative of coordinated smuggling activities.

Finding 1.2 Steganography and Concealed Documents

A steganographic image file named BNE.png was recovered from Steve's device in the following directory: "/img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Misc/BNE.png"

BNE.png initially appeared to be a standard image of Brisbane. Further investigation with Autopsy's file content viewer revealed that the file had signs associated with image-based steganography due to its larger file size compared the original image (1540752698-brisbane.jpg). When exposed to steganographic extraction, the photograph revealed a hidden image of a suitcase bundle suited for narcotics storage.

The recovered hidden image matches a file located in plain view on John Fredricksen's device (package.jpg), suggesting that John embedded the original image into 1540752698-brisbane.jpg before forwarding it to Steve. This supports the presence of a covert communication channel used to exchange operational material relating to the smuggling operation.

File Name	BNE.png
Analysis	Steganography from John to Steve
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Misc/BNE.png
File Size	4376793 bytes
Created Time	2019-02-01 08:13:19 SGT
Accessed Time	2019-02-02 10:31:06 SGT
Modified Time	2019-02-01 08:13:20 SGT
SHA-256 Hash	8a9a7dc6d6322d1ac8b089f1e921cf45ee970352964aaf6f8d4f649a3f554d3e

A second file named secret was also located in: "/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/secret"

Preliminary inspection suggests that this file also contains embedded or encrypted content. While the decryption key has been recovered, full examination and content extraction require further analysis to maintain evidentiary integrity. The existence of this file indicates an additional layer of concealment beyond standard steganographic practices.

File Name	secret
Analysis	TBD
File Path	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/secret
File Size	0
Created Time	-
Accessed Time	-
Modified Time	-
MD5 Hash	d41d8cd98f00b204e9800998ecf8427e

The application Image Steganography.exe was recovered at: "/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/CPascoe/ImageSteganography/ProgData/Image Steganography.exe" This executable corresponds to a common lightweight steganography tool. Although no application logs were located during the examination, the presence of the executable, combined with concealed images found on the system, demonstrates active use of digital obfuscation techniques.

The presence of a steganography application, combined with concealed images containing narcotics-related material, confirms deliberate attempts to hide operational content from detection. The correlation between the hidden image on Steve's device and its openly stored counterpart on John's device provides strong evidence of coordinated communication and shared knowledge of concealment techniques.

File Name	Image Steganography.exe
Analysis	Steganography Tool
File Path	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/CPascoe/Image Steganography/ProgData/Image Steganography.exe
File Size	204289
Created Time	2019-02-01 08:16:49 SGT
Accessed Time	2019-02-02 10:38:08 SGT
Modified Time	2019-02-01 08:16:48 SGT
MD5 Hash	1cb5862a24d1b8ec704559099c05dd76

Finding 1.3 Drug-Related Images and Deleted Materials

In Steve's device, two drug-related photos were discovered in the Pictures directory. These images depict cash bundles and narcotics that are consistent with methamphetamine. The system's Recycle Bin and orphaned file clusters had more erased versions of these images, indicating that they were removed on purpose.

Recovered Images (Visible Files)

File Name	price-meth-bust-4.jpg
Analysis	Possible Evidence of Drug Dealing by Steve
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/price-meth-bust-4.jpg
File Size	118136 bytes
Created Time	2019-01-31 10:58:22 SGT
Accessed Time	2019-01-31 11:04:13 SGT
Modified Time	2019-02-01 10:48:41 SGT
MD5 Hash	480da4ff5734c6d20787af08fc9da46e

File Name	620x349.jpg
Analysis	Product
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/620x349.jpg
File Size	19342 bytes
Created Time	2019-01-31 10:57:04 SGT
Accessed Time	2019-01-31 11:04:13 SGT
Modified Time	2019-02-01 10:48:41 SGT
MD5 Hash	60398a8ec59753ce5aadb99a566844d7

Recovered Deleted Images

File Name	$RA3IE5E.jpg
Analysis	Potential Proof of Steve Attempting to Destroy Evidence
File Location	/img_Narcos-1.001/vol_vol7/$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/$RA3IE5E.jpg
File Size	118136 bytes
Created Time	2019-01-31 10:58:22 SGT
Accessed Time	2019-01-31 11:04:13 SGT
Modified Time	2019-01-31 10:58:22 SGT
MD5 Hash	480da4ff5734c6d20787af08fc9da46e

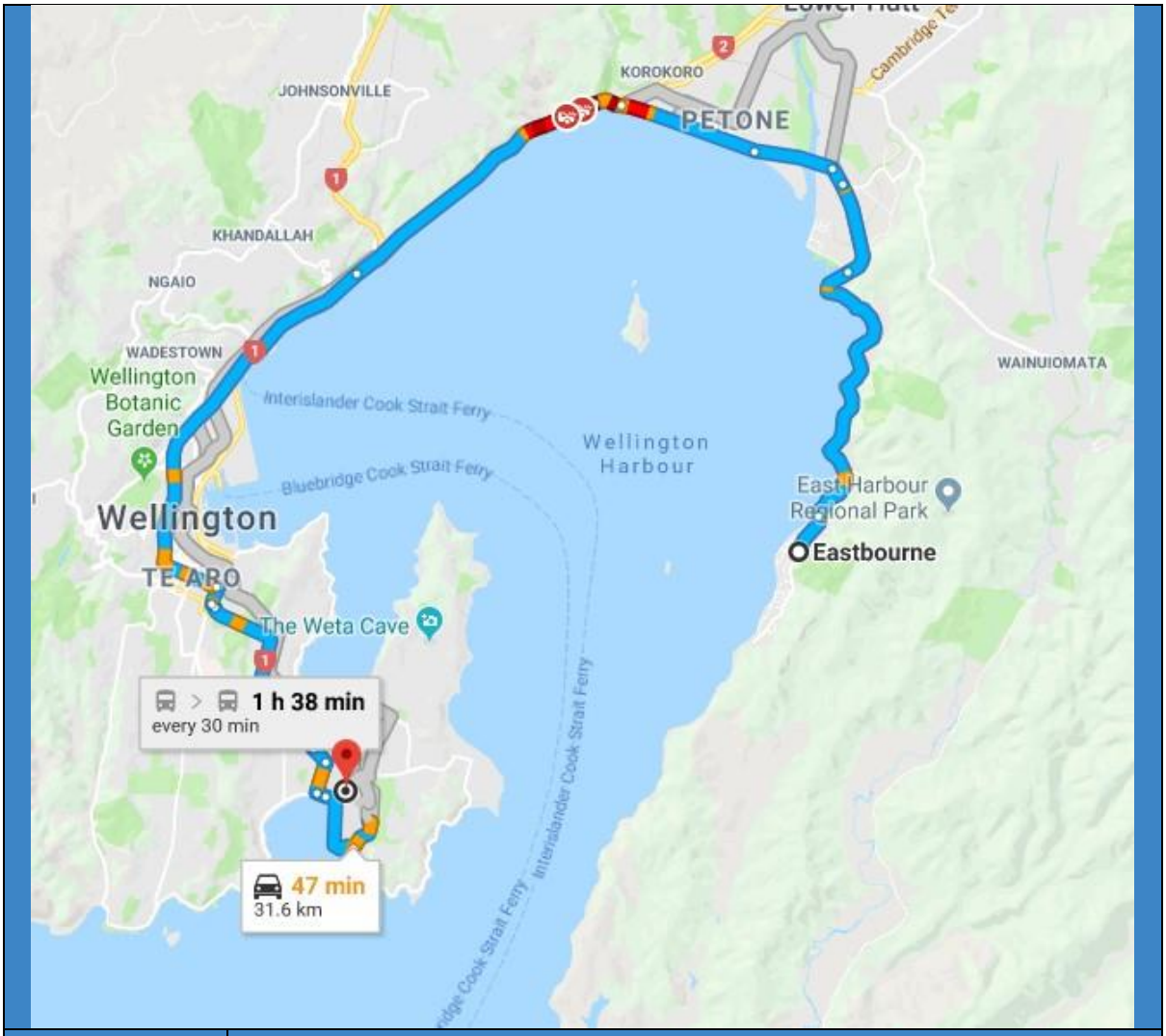
File Name	$RIIK1AS.jpg
Analysis	Potential Proof of Steve Attempting to Destroy Evidence
File Location	/img_Narcos-1.001/vol_vol7/$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/$RIIK1AS.jpg
File Size	19342 bytes
Created Time	2019-01-31 10:57:04 SGT
Accessed Time	2019-01-31 11:04:13 SGT
Modified Time	2019-01-31 10:57:06 SGT
MD5 Hash	60398a8ec59753ce5aadb99a566844d7

Autopsy also discovered orphaned clusters storing remains of the same images, indicating that many instances persisted throughout the filesystem. Copies were found in the Pictures, OrphanFiles, and Recycle Bin directories. This distribution suggests that the files be accessed or manipulated multiple times before being deleted.

Multiple copies of drug-related photographs, as well as matching deleted versions in the Recycle Bin, demonstrate that the material was regularly used and that attempts were made to erase evidence from the device. The images are consistent with drug-related content indicated in the Discord logs, despite no metadata being found that links them directly to the other suspects' devices (see Finding 1.1). The absence of significant EXIF metadata does not reduce evidential value. In drug-trafficking instances, file content and user activity patterns are usually more important than device-origin signals.

Finding 1.4 Operational and Travel Intelligence

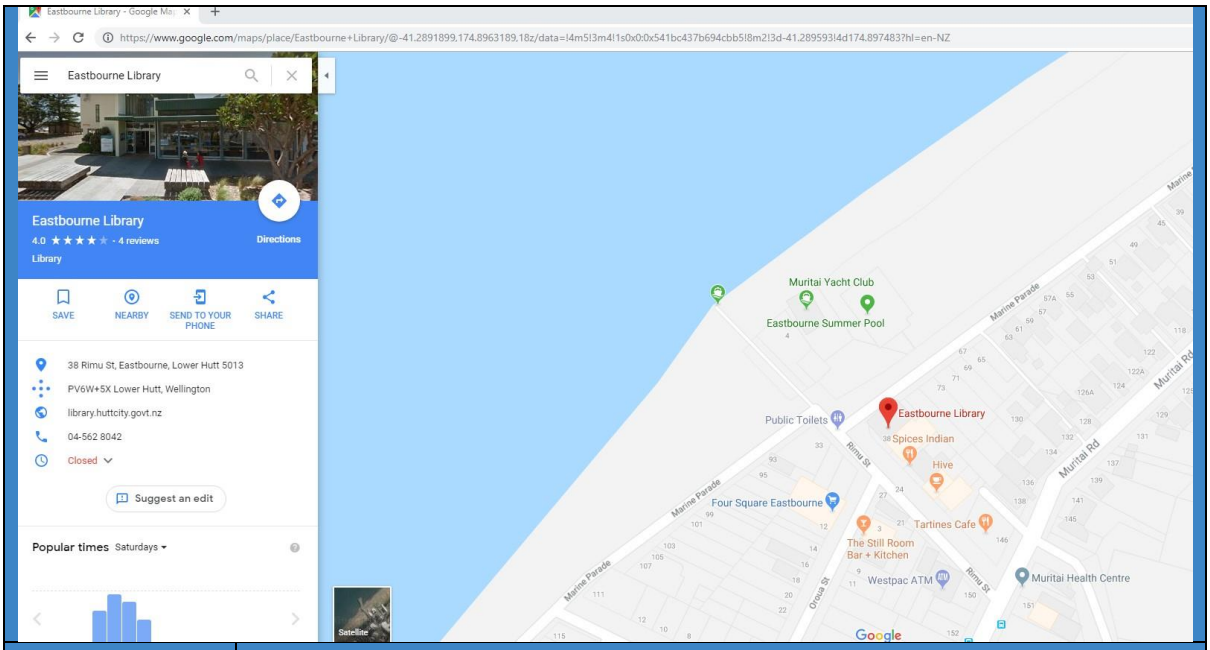
Multiple photographs obtained from Steve's device show detailed operational planning for the transit, delivery, and distribution of drugs. These files were found in the user's Documents/Misc directory and include map routes, drop-off site photos, airline itinerary details, and probable distribution channels. All files remained intact and were not flagged as destroyed.



File Name	airport crystals.jpg
Analysis	Route to Dropoff
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/airport crystals.jpg
File Size	77643 bytes
Created Time	2019-01-31 05:25:18 SGT
Accessed Time	2019-01-31 11:04:13 SGT
Modified Time	2019-01-31 05:25:18 SGT
MD5 Hash	f938f92d9dbaeda2e92246deeebe830a

The image above shows a Google Maps journey from an undetermined starting point to Wellington Airport, then to the Eastbourne Library—a site mentioned in Discord discussions recovered from the device (see Finding 1.1).

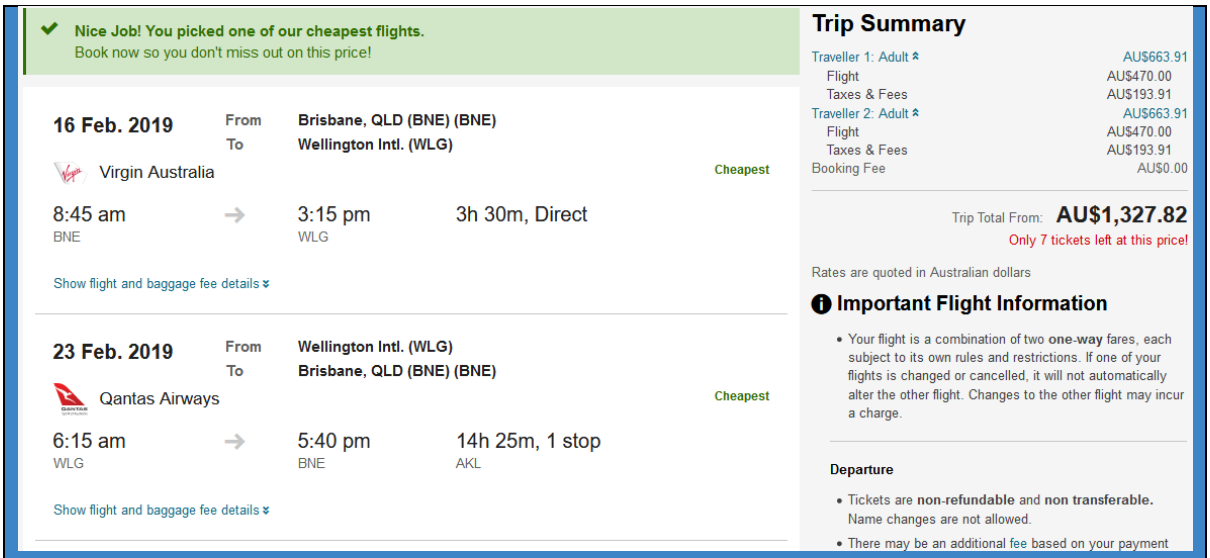
The existence of this route strongly corresponds to instructions contained in the communication logs about where the narcotics were to be delivered. Although the initial starting place is unknown, the explicit route to Eastbourne Library supports the investigative narrative that this location was intended as the primary delivery point.



File Name	dropoff.jpg
Analysis	Dropoff Location
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/dropoff.jpg
File Size	177888 bytes
Created Time	2019-02-02 09:06:05 SGT
Accessed Time	2019-02-02 10:31:06 SGT
Modified Time	2019-02-02 09:06:06 SGT
MD5 Hash	7feb763bebe42c3a3464c9ee3bb1d3e8

The image above depicts a Google Maps image showing the exterior and map view of the Eastbourne Library, the same address given by suspect Jane Esteban during her interview.

The information confirms that Steve had visual references to the scheduled meeting site, which corroborates both Jane's Discord discussions and real-world statements. This shows that Steve was involved in coordinating or monitoring the physical logistics of the trade.



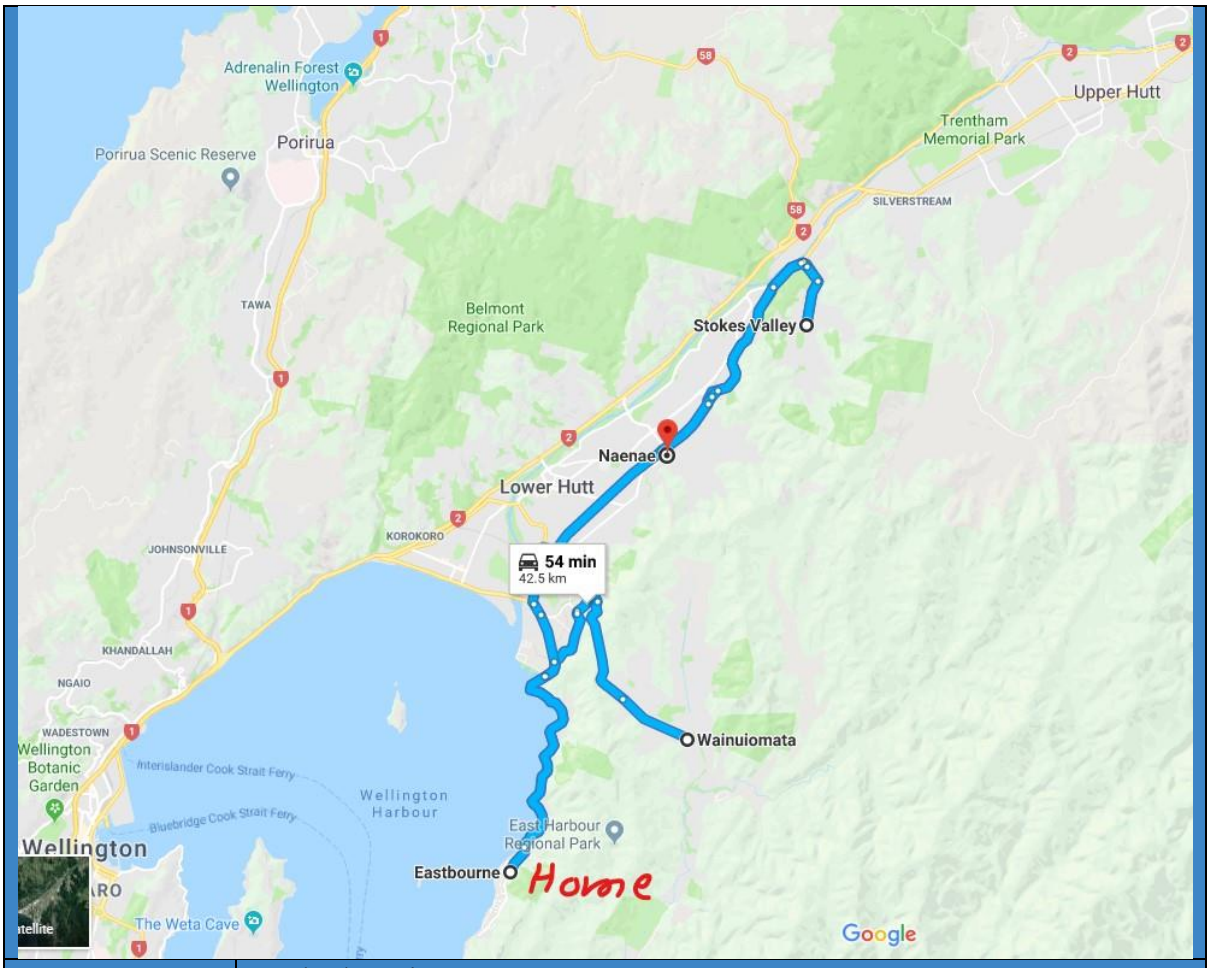
File Name	flightbookings.PNG
Analysis	Flight Detail for John and Jane to New Zealand
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/flightbookings.PNG
File Size	52971 bytes
Created Time	2019-02-02 10:28:44 SGT
Accessed Time	2019-02-02 10:31:06 SGT
Modified Time	2019-02-02 10:28:45 SGT
MD5 Hash	d1b21a1cddcb3494d637f2424cc5f0f1

An image containing airline booking information summarising travel itineraries from Brisbane to Wellington and Wellington back to Brisbane:

- Brisbane to Wellington: 16 February 2019, 08:45–15:15
- Wellington to Brisbane: 23 February 2019, 06:15–17:40

These dates correspond to the same timeframe as the Discord communications (30 January - 2 February 2019), which discussed the acquisition and delivery of drugs.

The flight itinerary implies that the suspects' overseas journey was well planned and coordinated. Although no direct link to the other suspects' devices was found, the timing and directional flow of traffic were consistent with the suspected smuggling operation



File Name	Method run.jpg
Analysis	Potential Location and Routes of Clients
File Location	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/Method run.jpg
File Size	145213 bytes
Created Time	2019-01-31 05:21:24 SGT
Accessed TIme	2019-01-31 11:04:13 SGT
Modified Time	2019-01-31 05:22:24 SGT
MD5 Hash	69b225bc0ef5a0c003f1e2a6646aaf92

The image above illustrates an image of Google Maps of a potential distribution path. The lines appear to connect different regions around Wellington from Eastbourne Library, the primary exchange location, indicating that narcotics were intended to be moved after the initial delivery.

The presence of mapped distribution channels suggests that planning extends beyond the initial drop-off. This shows that Steve may have been in charge of planning distribution or liaising with clients and parties not included in the provided dataset.

The collection of route maps, drop-off site photos, airline itineraries, and distribution-route visualisations provides compelling circumstantial proof of operational planning. These artefacts confirm the content of Discord discussions and are consistent with comments made during real-world questioning with the accused.

Together, these files demonstrate:

- knowledge of delivery locations
- coordinated travel scheduling
- preparation for distribution activities within Wellington
- possession of logistical intelligence related to the smuggling operation

Although no direct cross-device duplication of these photographs was found, the similarities between these files and other recovered evidence support the notion that Steve was involved in the strategic planning phase of the narcotics smuggling and distribution scheme.

Finding 1.5 Anti-Forensics

An examination of Steve's device indicated the existence of anti-forensic tools and virtualisation components, indicating an understanding of forensic countermeasures and an attempt to reduce digital traceability. The important discoveries include CCleaner installation, related log fragments, and a structured VMware program directory.

Finding 1.5.1 Anti-Forensics (CCleaner)

File Name	CCleaner.exe
Analysis	Potential Software Used to Reduce Traceability
File Path	/img_Narcos-1.001/vol_vol7/Program Files/CCleaner/CCleaner.exe
File Size	14679256
Created Time	2019-01-10 18:01:44 SGT
Accessed Time	2019-02-02 10:38:42 SGT
Modified Time	2019-01-10 18:01:44 SGT
MD5 Hash	f0f196fd71b9c7d65929861dd1bcfdcf

The presence of CCleaner indicates that a privacy-focused cleanup application has been installed on Steve's machine. CCleaner is widely used to clean up browser history, cache files, recent file listings, and system traces, all of which are anti-forensic acts.

Recovered CCleaner-Related Artifacts

Autopsy revealed multiple JavaScript files in the $OrphanFiles directory that referred to CCleaner activities or settings states:

File Name	dialog.min[1].js
Analysis	Logs of Enabling Smart Cleaning
File Path	/img_Narcos-1.001/vol_vol7/$OrphanFiles/dialog.min[1].js
File Size	12139
Created Time	2019-01-29 12:29:16 SGT
Accessed Time	2019-01-29 12:29:16 SGT
Modified Time	2019-01-29 12:29:16 SGT
MD5 Hash	37840a37213b7bf578ddb4d7ec214911

Contains indicators that smart cleaning is enabled. This suggests that automatic browser cleaning was enabled, and system notifications for identifying junk files were disabled. These settings would hide visible evidence of browsing activity and user interaction.

File Name	hqdefaultPBGJZ8A2.jpg
Analysis	AntiVirus and AntiSpyware Log of Update Notifications
File Path	/img_Narcos-1.001/vol_vol7/$OrphanFiles/hqdefaultPBGJZ8A2.jpg
File Size	7093
Created Time	2019-01-29 11:13:26 SGT
Accessed Time	2019-01-29 11:13:26 SGT
Modified Time	2019-01-29 11:13:26 SGT
MD5 Hash	f2bed1c8f610dd8b57e8ca65110a2acd

This file appears to be part of system logs relating to antivirus and antispyware update notifications. Its presence in the orphaned directory suggests CCleaner or another cleaning process may have removed the original logs, leaving only partial remnants.

File Name	jquery.themepunch.revolution.min[1].js
File Path	/img_Narcos-1.001/vol_vol7/$OrphanFiles/jquery.themepunch.revolution.min[1].js
File Size	64971
Created Time	2019-01-29 12:29:15 SGT
Accessed Time	2019-01-29 12:29:15 SGT
Modified Time	2019-01-29 12:29:15 SGT
MD5 Hash	aa3c02dc2423e17d68a083b2b7cab4ab

Covers generic references to CCleaner cleaning procedures. While not particular, its presence in an orphaned directory suggests that earlier cleaning cycles have erased the original file structure..

CCleaner.log and SetupLog.txt, two expected CCleaner artefacts, were not located. Their absence is consistent with either cleaning actions occurring prior to disk imaging or CCleaner removing its own logs.

Steve may have attempted to limit recoverable browser and system activity, as demonstrated by the presence of CCleaner, orphaned log fragments, and smart cleaning configuration traces. Although CCleaner is not inherently harmful, its use in a drug distribution case implies the intentional destruction of potential digital evidence.

Finding 1.5.2 Anti-Forensics (VMware)

File Name	VMware
Analysis	Directory of VMware
File Path	/img_Narcos-1.001/vol_vol7/Program Files/VMware
File Size	264
Created Time	2019-01-29 03:38:02 SGT
Accessed Time	2019-02-02 10:38:19 SGT
Modified Time	2019-01-29 03:38:02 SGT
MD5 Hash	-

The directory contained several DLL files, executables, and configuration components typically found in a VMware Tools installation. These files prove that Steve's device supported virtual machine environments.

There was no direct evidence of:

- virtual machine disk files (.vmdk, .vdi, .vhd),
- snapshot or saved-state files, or
- recent VMware execution traces.

As a result, while VMware Tools was installed, there is no conclusive evidence that a virtual machine was being used within the relevant time frame.

The installation of VMware Tools indicates that the system was either part of a virtualisation environment at some point or was ready to operate virtual machines. Virtual machines are occasionally used for sandboxing, isolating incriminating activity, or testing concealing strategies. Nevertheless, no incriminating files relating to VM usage were discovered on Steve's device.

The installation of CCleaner and its associated orphaned log artefacts are likely attempts to limit recoverable evidence of Steve's digital activity. This is consistent with anti-forensic techniques commonly employed to conceal user interactions, browser history, and operational planning data. Although VMware artefacts were present, no active virtual machine activity was detected.

Overall, these data suggest that Steve had the knowledge and resources to limit digital traceability, supporting the idea that he was engaging in deliberate concealment behaviours in the context of the narcotics-related activities apparent elsewhere on the device.

Finding 1.6 Summary

A thorough acquisition of artefacts from the forensic investigation of Steve Kowhai's device (Narcos-1.001) revealed that he was intimately involved in the planning, coordination, and concealment of drug-related operations. The evidence collected from the system includes communication logs, steganographic material, operational planning files, drug-related imagery, and signs of anti-forensic conduct.

Discord communications obtained from the device contained discussions about drug quantities, delivery instructions, and meeting spots. These messages made it evident that Steve was involved in transaction logistics and mentioned steganography. His involvement in covert communication channels is further substantiated by the availability of steganographic files, such as BNE.png, which contains a drug-related hidden image as well as another hidden file (secret). The existence of Image Steganography.exe promotes the active use of digital concealment techniques.

The Pictures, Recycle Bin, and orphaned file folders all contained drug-related photographs, both accessible and deleted, showing repeated interaction with the content and attempts to erase damning content. A variety of images depicting routes, drop-off locations, airline itineraries, and distribution patterns were also obtained from Steve's Documents directory. These data demonstrate a high level of operational planning because they exactly match the delivery locations specified in Discord discussions and confirm the facts discovered during the real investigation.

Anti-forensic artefacts were also found on the device. The presence of CCleaner, together with orphaned fragments referencing smart-cleaning and system cleanup activity, implies efforts to reduce recoverable digital traces. Although VMware Tools were installed, there was no indication of active use of virtual machines throughout the relevant time period.

Overall, the findings from Steve's device show his active engagement in directing the smuggling operation, technical skill to disguise information, and awareness of forensic countermeasures. These artefacts provide strong incriminating evidence, linking him to the narcotics trafficking scheme's planning, communication, and attempted concealment.

Finding 2 - Analysis of John’s Device (Narcos.2-001)

Windows artefacts for Narcos-2 (host JOHNFLAPTOP1, owner “JohnF” in Operating System Information) were examined to corroborate John’s role in receiving and handling the suspicious “Attachments-Important” and “Contact_Card” files, his use of ProtonMail and Discord, and potential transfer of data to removable media. OS Accounts shows a domain user johnf created on 29 January 2019 at 03:25:46 SGT, establishing him as the primary interactive user associated with later activity on this system.

Web Downloads artefacts provide clear evidence of John acquiring both the steganography tool and the key archive files. On 1 February 2019 at 07:00:16 SGT, Firefox downloads Image Steganography 1.5.2 Setup.exe to C:\Users\JohnF\Downloads\ from a SourceForge URL, mirroring Steve’s acquisition of a similar stego tool on Narcos-1. Earlier, Windows records alternate data stream (Zone.Identifier) entries under /Users/JohnF/Downloads/Attachments-Important, crucial to our method/secret, /Users/JohnF/Downloads/Attachments-Important, crucial to our method.zip and /Users/JohnF/Downloads/Contact_Card.zip. The ADS for Attachments-Important, crucial to our method.zip is linked to a blob:https://mail.protonmail.com/... URL, indicating the ZIP was downloaded directly from ProtonMail, while the ADS for Contact_Card.zip references https://cdn.discordapp.com/attachments/.../Contact_Card.zip, tying that archive to Discord. These Windows records show that John’s laptop received both the “Attachments-Important” package and the “Contact_Card” archive from external services (ProtonMail and Discord) and that the folder contained a file named secret with an Internet zone marker.

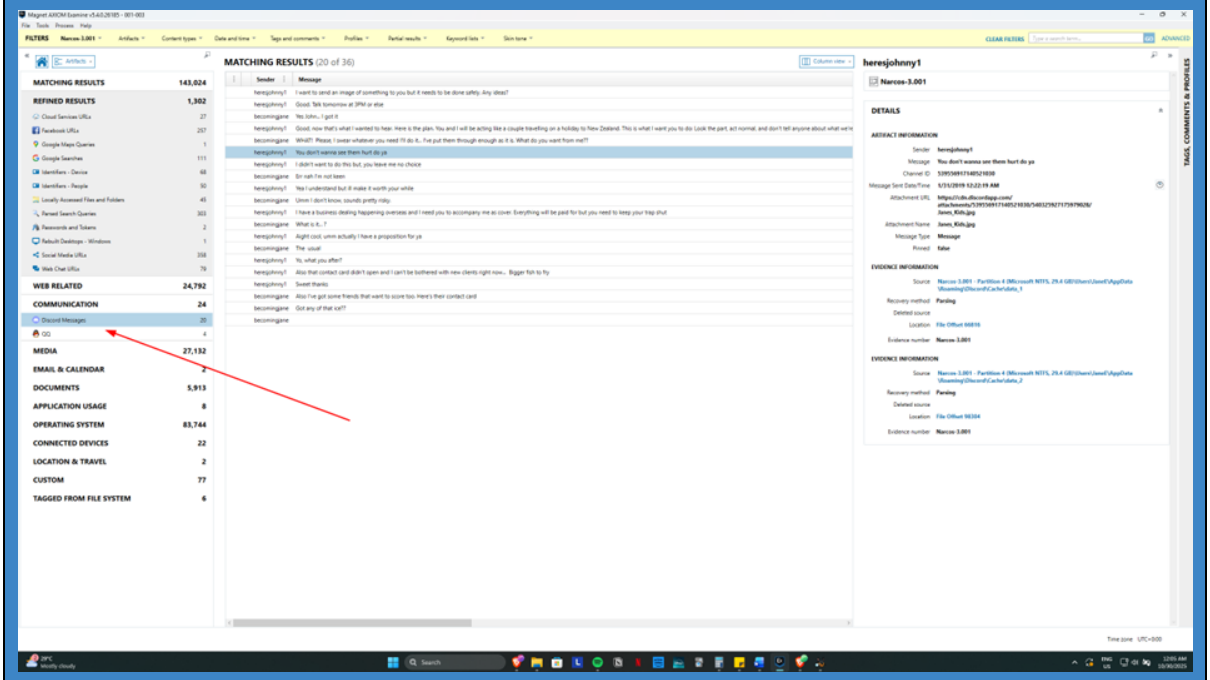
Web History and Shell Bag artefacts then corroborate that John interacted with the downloaded content rather than merely storing it. Web History entries from WebCacheV01.dat show Microsoft Edge accessing file:///C:/Users/JohnF/Downloads/Attachments-Important at 30 January 2019 00:00:16 SGT under the JohnF profile. At almost the same time, Shell Bags parsed from UsrClass.dat record a folder view path My Computer\{088e3905-0323-4b02-9826-5d99428e115f}\Attachments-Important, crucial to our metho, with Last Write, Created, Modified and Accessed timestamps all at 30 January 2019 00:00:50 SGT. This combination of Edge history and Shell Bag entries confirms that John opened the Attachments-Important, crucial to our method folder in Explorer and browsed its contents shortly after the ProtonMail-delivered ZIP arrived. Recent Documents further supports this: it contains shortcuts Attachments-Important, crucial to our method.lnk and Attachments-Important, crucial to our method.zip.lnk pointing to C:\Users\JohnF\Downloads\Attachments-Important, crucial to our method.zip, with one LNK showing a Date Accessed of 30 January 2019 08:00:16 SGT. Taken together, Windows Web Downloads, Web History, Shell Bags and Recent Documents artefacts independently demonstrate that John both downloaded and opened the “Attachments-Important” archive and its extracted folder.

Additional Windows artefacts reinforce John's use of communication channels and removable media. Web History shows multiple accesses to https://protonmail.com/, the ProtonMail login page and https://mail.protonmail.com/inbox, including a Firefox history entry titled "(4) Inbox | heresjohnny1@protonmail.com | ProtonMail" at 1 February 2019 08:05:24 SGT, confirming active use of the ProtonMail account heresjohnny1@protonmail.com on this host. Earlier, John searches for and visits https://discordapp.com/ around 29 January 2019 04:20 SGT, which aligns with the Discord-sourced Contact_Card.zip recorded in Web Downloads. Recent Documents also records activity on an external volume E:\ through Local Disk (E).lnk and Memo Things.lnk pointing to E:\Memo Things.odt, both accessed on 30 January 2019 08:04:22 SGT. USB Device Attached artefacts for Narcos-2 show a Western Digital "Elements Portable (WDBUZG)" USB device with Device ID 57584D3145373444574D314E connected at 2 February 2019 10:56:41 SGT; the same device identifier is also observed on Steve's machine (Narcos-1) on 1 February 2019, suggesting a shared removable drive used across both systems. Overall, the Windows artefacts for Narcos-2 strongly corroborate that John, using the johnf account, logged into ProtonMail and Discord, downloaded the "Attachments-Important" and "Contact_Card" archives, opened the "Attachments-Important" folder and its contents (including a file named secret), and had access to an external drive that also appears on Steve's system, providing a plausible channel for moving or exchanging concealed data between the two suspects

Finding 2.1 Communications

The primary communication artefacts on John's device are Discord LevelDB storage files located under C:\Users\JohnF\AppData\Roaming\Discord\Local Storage\leveldb\. These include 000007.ldb, created on 29 January 2019 at 10:33:31 SGT, and the later pair 000013.log and 000014.ldb, created on 31 January 2019 at 08:26:18 SGT and last accessed on 2 February 2019 shortly before seizure. The timestamps show that Discord was installed and actively used across the critical period of 29–31 January, with artefacts still being accessed up to 2 February, indicating recent Discord activity by user JohnF in the lead-up to his arrest.

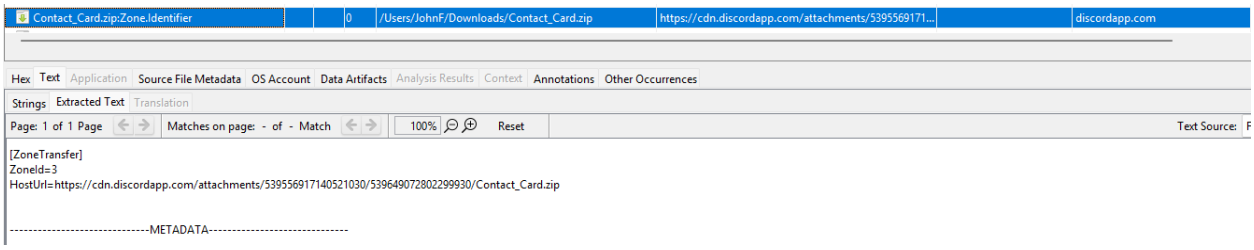
File Name	000007.ldb
Analysis	Discord messages
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000007.ldb
File Size	8614
Created Time	2019-01-29 10:33:31 SGT
Accessed Time	2019-02-02 11:01:29 SGT
Modified Time	2019-01-29 10:33:31 SGT
MD5 Hash	32ee12999e8690129d53b1fbacb49fea



File Name	000013.log
Analysis	Discord messages
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000013.log
File Size	24834
Created Time	2019-01-31 08:26:18 SGT
Accessed Time	2019-02-02 11:04:00 SGT
Modified Time	2019-02-02 09:57:37 SGT
MD5 Hash	fec64fad23da36b18ae532c99c81933a

File Name	000014.ldb
Analysis	Discord messages
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000014.ldb
File Size	10526
Created Time	2019-01-31 08:26:18 SGT
Accessed Time	2019-02-02 11:01:29 SGT
Modified Time	2019-01-31 08:26:18 SGT
MD5 Hash	857d3ac75d8345f27e45cf5bf4338713

These .log and .ldb files form part of Discord's LevelDB-based local cache, which records message content, channel metadata, timestamps and user interaction for the logged-in account. Although Magnet AXIOM did not successfully parse human-readable conversations directly from John's LevelDB store, the presence and timing of these files align with John's known use of Discord recorded elsewhere in the Windows artefacts. Web history shows that John browsed to https://discordapp.com/ around 29 January 2019, and Web Downloads records Contact_Card.zip being retrieved from a Discord CDN URL, confirming that Discord was an active channel for receiving operational files on this host.



John's device also evidences use of ProtonMail as a parallel communication channel. Browser artefacts show repeated visits to https://protonmail.com, the login page and the inbox, including a Firefox history entry titled "(4) Inbox | heresjohnny1@protonmail.com | ProtonMail" on 1 February 2019 at 08:05:24 SGT. This ties the laptop to the ProtonMail account heresjohnny1@protonmail.com. Alternate Data Stream (Zone.Identifier) entries on Attachments-Important, crucial to our method.zip link its download to a blob:https://mail.protonmail.com/... URL, demonstrating that John not only logged into ProtonMail but also received and saved the "Attachments-Important" archive via this encrypted email platform.

File Name	(4) Inbox | heresjohnny1@protonmail.com | ProtonMail
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite
File Size	10526
Created Time	2019-01-29 04:09:54 SGT
Accessed Time	2019-02-02 10:47:52 SGT
Modified Time	2019-02-02 10:47:52 SGT
MD5 Hash	50a698fff857f00065e7b6b0ceee2188

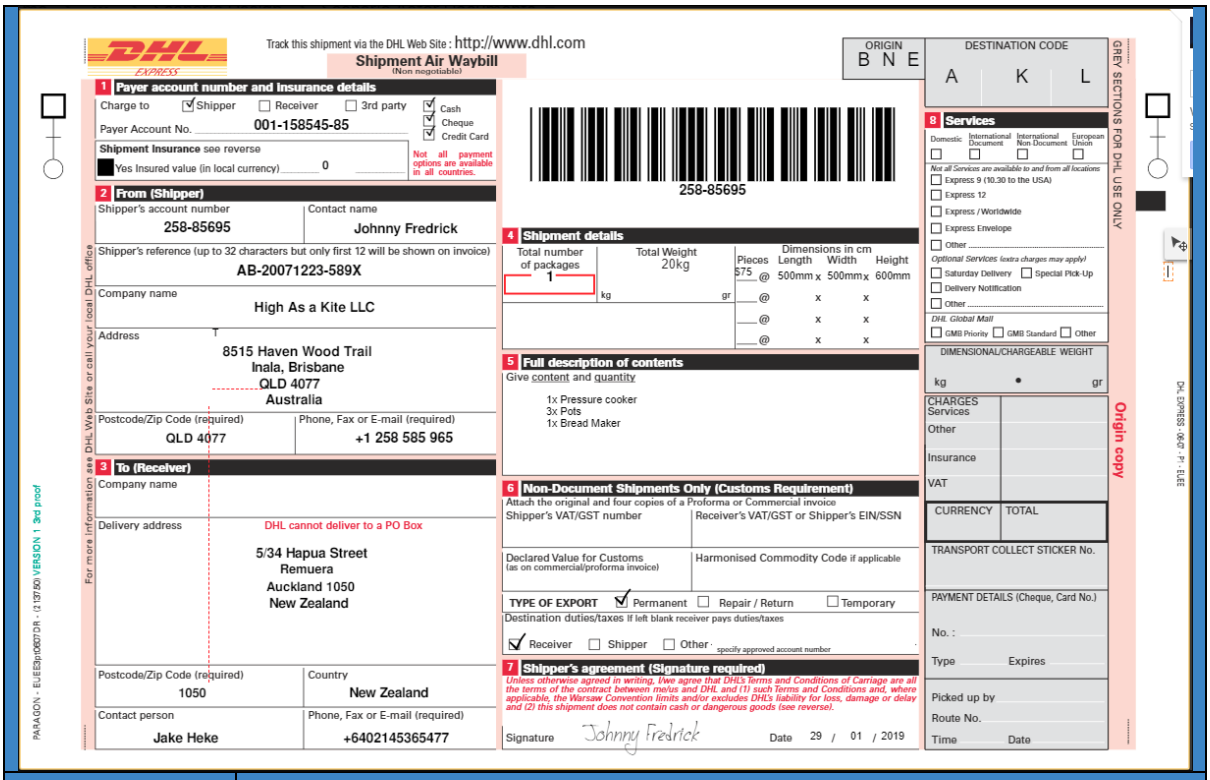
Taken together, the Discord LevelDB artefacts, ProtonMail browsing history and Zone.Identifier records show that John used at least two covert-friendly communication channels: Discord for receiving Contact_Card.zip and ProtonMail for receiving Attachments-Important, crucial to our method.zip. While the raw contents of John's Discord logs on this image were not directly decoded, their presence, timestamps and locations, when correlated with the ProtonMail and archive-download artefacts, support the conclusion that John's device functioned as an active endpoint for both real-time chat and encrypted email coordination in the narcotics operation.

Finding 2.2 Drug-Related Operational Documents

The file clients.ods is located in John's Business folder, it is a small OpenDocument spreadsheet created on 29 January 2019 at 04:26:09 SGT and last modified at 12:02:47 SGT the same day, with an access time of 30 January 2019 at 08:04:14 SGT. The spreadsheet contains a structured table with the columns Name, Location, Product, Amount, Delivery, listing multiple "clients" such as Ricky Ross, Frank Lucas, Chris Coke, Manuel Noriega, Joaquin Guzman, Al Capone and others, each associated with specific locations (e.g. Los Angeles, New York, Panama, Guadalajara), drug slang terms (e.g. "Mama Coca", "Ferry Dust", "Smack", "China White") and regular shipment quantities (typically 10–25 kg on monthly or quarterly schedules). One entry explicitly lists "Jane Esteban, Brisbane, Australia, Uppers, 1 gram, On demand", which stands out as a retail-sized quantity and an "on demand" schedule rather than a bulk shipment, indicating that Jane is recorded differently from the major wholesale-style entries.

File Name	clients.ods
Analysis	Client list
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/clients.ods
File Size	15077
Created Time	2019-01-29 04:26:09 SGT
Accessed Time	2019-01-30 08:04:14 SGT
Modified Time	2019-01-29 12:02:47 SGT
MD5 Hash	db804b47095901e20d6666b1e9a56003

The structure and content of clients.ods strongly suggest that it is being used as a drug client and shipment planning ledger rather than a legitimate business contact list. The use of well-known historical and contemporary narcotics trafficker names (e.g. "Frank Lucas", "Joaquin Guzman", "Al Capone") is consistent with the use of code names or personas rather than genuine customer identities, while the pairing of those names with specific locations, products and regular bulk quantities indicates an attempt to track ongoing supply commitments. The presence of Jane's details in the same table, but with a small "on demand" quantity, ties her persona directly into John's operational planning documents, consistent with her being treated as a lower-volume client or special case within the distribution network.



DHL EXPRESS

Track this shipment via the DHL Web Site : http://www.dhl.com

Shipment Air Waybill
(Non negotiable)

ORIGIN B N E

DESTINATION CODE A K L

1 Payer account number and Insurance details

Charge to ☑ Shipper ☐ Receiver ☐ 3rd party ☑ Cash ☑ Cheque ☑ Credit Card

Payer Account No. 001-158545-85

Shipment Insurance see reverse

Yes Insured value (in local currency) 0

Not all payment options are available in all countries.

258-85695

2 From (Shipper)

Shipper's account number: 258-85695

Contact name: Johnny Fredrick

Shipper's reference (up to 32 characters but only first 12 will be shown on invoice): AB-20071223-589X

Company name: High As a Kite LLC

Address: 8515 Haven Wood Trail, Inala, Brisbane, QLD 4077, Australia

Postcode/Zip Code (required): QLD 4077

Phone, Fax or E-mail (required): +1 258 585 965

3 To (Receiver)

Company name

Delivery address: 5/34 Hapua Street, Remuera, Auckland 1050, New Zealand

DHL cannot deliver to a PO Box

Postcode/Zip Code (required): 1050

Country: New Zealand

Contact person: Jake Heke

Phone, Fax or E-mail (required): +6402145365477

4 Shipment details

Total number of packages: 1

Total Weight: 20kg

Pieces: 575 @ Dimensions in cm: Length 500mm x Width 500mm x Height 600mm

5 Full description of contents

Give content and quantity

1x Pressure cooker
3x Pots
1x Bread Maker

6 Non-Document Shipments Only (Customs Requirement)

Attach the original and four copies of a Proforma or Commercial invoice

Shipper's VAT/GST number

Receiver's VAT/GST or Shipper's EIN/SSN

Declared Value for Customs (as on commercial/proforma invoice)

Harmonised Commodity Code if applicable

TYPE OF EXPORT ☑ Permanent ☐ Repair / Return ☐ Temporary

Destination duties/taxes If left blank receiver pays duties/taxes

☑ Receiver ☐ Shipper ☐ Other

7 Shipper's agreement (Signature required)

Unless otherwise agreed in writing, I/we agree that DHL's Terms and Conditions of Carriage are all the terms of the contract between me/us and DHL and (1) such Terms and Conditions and, where applicable, the Warsaw Convention limits and/or excludes DHL's liability for loss, damage or delay and (2) this shipment does not contain cash or dangerous goods (see reverse).

Signature: Johnny Fredrick

Date 29 / 01 / 2019

8 Services

Domestic | International Document | International Non Document | European Union

Not all Services are available to and from all locations

☐ Express 9 (10.30 to the USA)
☐ Express 12
☐ Express / Worldwide
☐ Express Envelope
☐ Other

Optional Services (extra charges may apply)

☐ Saturday Delivery ☐ Special Pick-Up
☐ Delivery Notification
☐ Other

DHL Global Mail

☐ GMB Priority ☐ GMB Standard ☐ Other

DIMENSIONAL/CHARGEABLE WEIGHT kg gr

CHARGES Services | Other | Insurance | VAT

CURRENCY | TOTAL

TRANSPORT COLLECT STICKER No.

PAYMENT DETAILS (Cheque, Card No.)

No. :

Type Expires

Picked up by

Route No.

Time Date

GREY SECTIONS FOR DHL USE ONLY

Origin copy

For more information see DHL Web Site or call your local DHL office

PARAGON - EUEE3jn1060/7DR - (1197/50) VERSION 1 3rd proof

DHL EXPRESS - 06/07 - P - EUEE

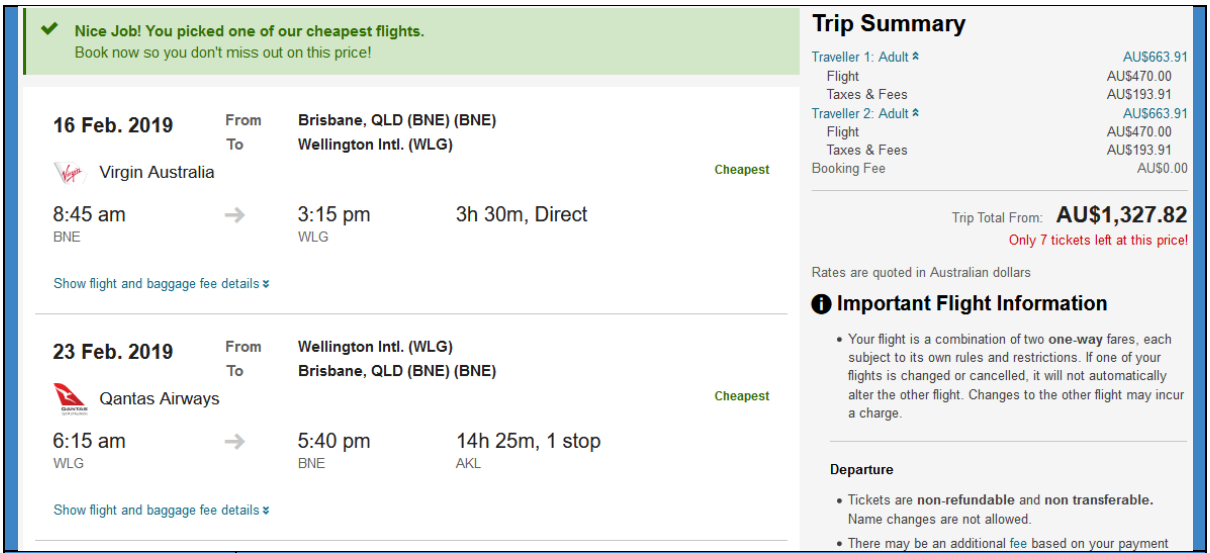
File Name	shipping.PNG
Analysis	Shipment detail of potential narcotics
File Size	189265 bytes
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/shipping.PNG
Created Time	2019-01-23 09:18:28 SGT
Accessed Time	2019-01-29 11:58:47 SGT
Modified Time	2019-01-23 09:18:28 SGT
MD5 Hash	4fdf3dd6bac6ba99f6eafcedd9a64efa

A second file, shipping.PNG, is also stored in the same Business directory. It is a PNG image created on 23 January 2019 at 09:18:28 SGT and last accessed on 29 January 2019 at 11:58:47 SGT, shortly before clients.ods was last modified at 12:02:47 SGT.

The file's name, location under Documents/Business, and access timing indicate that it is a business-related diagram or document used as a visual reference for shipping or logistics planning. The close temporal correlation between John opening shipping.PNG and then modifying clients.ods suggests that he consulted this shipping image while updating the client and delivery schedule spreadsheet, implying that the two artefacts are used together as part of the same drug distribution planning workflow.

These documents place John in an organiser/coordinator role within the network, responsible for tracking customers, products, quantities and delivery frequencies rather than acting solely as an end-user or courier.

Finding 2.3 Flight and Travel Information



File Name	Steve K.PNG
Analysis	Flight Details of John and Jane
File Size	52971 bytes
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/Steve K.PNG
Created Time	2019-02-02 06:43:20 SGT
Accessed Time	2019-02-02 09:57:27 SGT
Modified Time	2019-02-02 06:43:21 SGT
MD5 Hash	d1b21a1cddcb3494d637f2424cc5f0f1

This key travel-related artefact on John's device is the image file Steve K.PNG, stored in his Business directory. Hash analysis shows that Steve K.PNG on John's laptop shares the same MD5 hash (d1b21a1cddcb3494d637f2424cc5f0f1) and file size as flightbookings.PNG recovered from Steve's device, which is stored under C:\Users\Steve\Documents\Misc\flightbookings.PNG. On Steve's machine, this image contains airline booking details summarising a return trip from Brisbane to Wellington and back for the period 16–23 February 2019. The outbound leg is from Brisbane to Wellington on 16 February 2019 (08:45–15:15), and the return leg is from Wellington to Brisbane on 23 February 2019 (06:15–17:40); these dates fall shortly after the Discord discussions and planning activity observed between 30 January and 2 February 2019. Because the two files are bit-for-bit identical, Steve K.PNG on John's laptop can be reliably interpreted as a copy of the same flight booking itinerary for John and Jane's New Zealand trip.

The presence of this flight itinerary image under John's Business directory, combined with its late creation and access times on 2 February 2019, shows that John was actively viewing and retaining the booking information in the final hours of the investigation period. When correlated

with the drug client ledger (clients.ods), shipping documentation (shipping.PNG), and the communications planning the Eastbourne Library handover, this artefact supports the conclusion that John was not only travelling but also engaged in deliberate planning and review of the smuggling trip's logistics. In effect, Steve K.PNG links John's device directly to the same travel plan documented on Steve's computer, reinforcing that both suspects were working to a coordinated itinerary for the Brisbane–Wellington run rather than undertaking unrelated personal travel.

Finding 2.4 Suspicious Files, Steganography and Concealment

Analysis of John's device (Narcos-2.001) shows that he employed multiple, layered concealment techniques: full-disk–style encryption via TrueCrypt to hide operational planning documents, and image steganography to conceal drug-related material inside seemingly benign pictures. Together with the presence of sensitive personal images, these artefacts indicate not only an attempt to protect information from detection but also to maintain leverage over other participants.

File Name	TrueCrypt.exe
File Path	/img_Narcos-2.001/vol_vol7/Program Files/TrueCrypt/TrueCrypt.exe
File Size	1516496
Created Time	2019-01-30 07:59:16 SGT
Accessed Time	2019-02-02 11:03:08 SGT
Modified Time	2019-01-30 07:59:16 SGT
MD5 Hash	fa8f08013422a4eb68072668b3a73293

Crystal Method

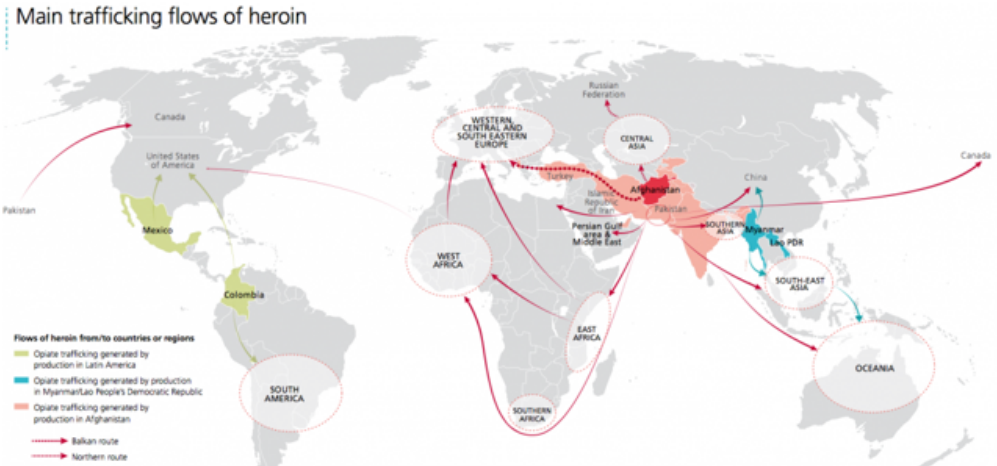
Destination – Wellington
Product – 1 kg crystal
Source – Brisbane (JF)

Wellington is part of the lower north island of New Zealand. It is accessable by road and sea. As tempting as the sea is, be aware that the coast of NZ may seem ideal, but everyone knows it's vulnerable as there have been several news reports about it.

The product is coming from Brisbane, Australia and as a result, will need to be delivered by sea or plane. Shipping the product would be ideal, the risk of detection is lower but would result in higher costs for a small amount of product. The product being delievered is only a test to check it's quality and will be the deciding factor in any decision to continue doing business with you. If the product is up to scratch, I will buy a larger supply next time. If we continue to do business it might be more practical to use shipping for future transactions.

This consignment will be delievered by plane, which will decrease travel time and lower the cost for all parties involved. Delievery by plane will bring along with it its own complications such as concealment, security screening/checks and drug dogs. I trust that you are aware of the risks and will take all necessary precautions.

Below is a map of drup flows from countires, could give us some info for later trades and where we can get more product from in the future.



File Name	E:\Memo Things.odt
Analysis	Hidden file in “secret” file
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Microsoft/Windows/Recent/AutomaticDestinations/5f7b5f1e01b83767.automaticDestinations-ms/Memo Things.odt.lnk
File Size	1516496
Created Time	2019-01-30 07:59:16 SGT
Accessed Time	2019-02-02 11:03:08 SGT
Modified Time	2019-01-30 07:59:16 SGT
MD5 Hash	18302ce1440a264fdd66ba51e1dc9ac2

File Name	Attachments-Important, crucial to our method.zip
Analysis	Location of “secret”
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Attachments-Important, crucial to our method.zip
File Size	5242990
Created Time	2019-01-30 08:00:16 SGT
Accessed Time	2019-01-30 08:00:50 SGT
Modified Time	2019-01-30 08:00:16 SGT
MD5 Hash	7774e97391fe79e3ca91141ca36412af

File Name	secret
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Attachments-Important, crucial to our method.zip/secret
File Size	5242880
Created Time	-
Accessed Time	-
Modified Time	2019-01-29 23:59:50 SGT
MD5 Hash	173094089256dfa0bbf2c682df261b5f

At the encryption layer, John possessed both the capability and the artefacts required to work with encrypted containers. TrueCrypt 7.1a was present on his system, and during laboratory analysis the file secret, which was originally recovered from Steve’s device and associated with John via Attachments-Important, crucial to our method.zip, which was was successfully mounted as a TrueCrypt volume using the passphrase “ilovediving”. Once mounted, the container was found to hold a single document, Memo Things.odt. That document, titled “Crystal Method”, outlines a 1 kg consignment of “crystal” from Brisbane to Wellington, identifies “JF” as the sender, and discusses transport modes, risk versus cost, and how this shipment will be used as a quality test for larger future orders. The fact that this memo exists only within an encrypted volume, and that John’s machine shows access to Memo Things.odt via a shortcut pointing to E:\Memo Things.odt on an attached USB drive, links him directly to the concealed planning of the Brisbane–Wellington shipment and demonstrates deliberate use of encryption to hide critical operational intelligence.

File Name	Image Steganography.exe
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/CPascoe/Image Steganography/ProgData/Image Steganography.exe
File Size	204289
Created Time	2019-02-01 07:00:47 SGT
Accessed Time	2019-02-02 11:03:08 SGT
Modified Time	2019-02-01 07:00:47 SGT
MD5 Hash	1cb5862a24d1b8ec704559099c05dd76

File Name	package.jpg
Analysis	Image hidden in “1540752698-brisbane.jpg”
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Local/Temp/vmware-JohnF/VMwareDnD/770cd05f/package.jpg
File Size	83781
Created Time	2019-02-01 07:02:24 SGT
Accessed Time	2019-02-01 07:04:37 SGT
Modified Time	2019-02-01 07:02:24 SGT
MD5 Hash	8ba9265a1563ff871a471c552bab0b67

At the steganography layer, John's device contains clear evidence of image-based concealment. In his Downloads and Program Files directories, installers and binaries for Image Steganography are present, and examination confirmed that Image Steganography.exe had been used with the passphrase "Elchapo2". The original photograph 1540752698-brisbane.jpg resides in his Pictures folder, representing a normal-sized cityscape image. A derivative file, BNE.png, is also present and is significantly larger in file size than the original JPEG, despite showing the same apparent scene when viewed normally. When BNE.png is processed through Image Steganography with the passphrase "Elchapo2", it reveals the embedded payload package.jpg. This concealed image depicts drug packaging consistent with the type of consignment discussed elsewhere in the case, transforming BNE.png from a simple Brisbane picture into a covert carrier for operational imagery. This confirms that John did not merely possess steganography tools; he used them to embed specific drug-related content inside an otherwise innocuous-looking file.

File Name	BNE.png
Analysis	Steganography
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Pictures/BNE.png
File Size	4376793 bytes
Created Time	2019-02-01 07:05:17 SGT
Accessed Time	2019-02-01 08:06:04 SGT
Modified Time	2019-02-01 07:05:17 SGT
MD5 Hash	dffa98bdd7371d1806adb09db27ba283

File Name	1540752698-brisbane.jpg
Analysis	Original Image for Steganography
File Location	/img_Narcos-2.001/vol_vol7/Users/JohnF/Pictures/1540752698-brisbane.jpg
File Size	274035 bytes
Created Time	2019-02-01 05:30:16 SGT
Accessed Time	2019-02-01 07:05:16 SGT
Modified Time	2019-02-01 05:30:17 SGT
SHA-256 Hash	8a9a7dc6d6322d1ac8b089f1e921cf45ee970352964aaf6f8d4f649a3f554d3e

File Name	Janes Kids.jpg
File Size	200136 bytes
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Local/Temp/vmware-JohnF/VMwareDnD/2cce6695/Janes Kids.jpg
Created Time	2019-01-23 10:29:41 SGT
Accessed Time	2019-01-31 08:20:56 SGT
Modified Time	2019-01-23 10:29:42 SGT
MD5 Hash	1b4ba1d430ab0f4a5f1698ec481a8f93

Further context for the coercive and concealment aspects comes from additional images associated with John's profile. Janes Kids.jpg, an image of Jane's children, and package.jpg both appear under a VMware drag-and-drop cache path (AppData\Local\Temp\vmware-JohnF\VMwareDnD\…), indicating they were being actively moved between host and virtual environments rather than passively stored. In the broader scenario, John is noted as having threatened Jane, and his possession of a personal family photograph alongside drug-package imagery is consistent with the ability to apply pressure or intimidation while simultaneously compartmentalising incriminating content inside encrypted containers and steganography carriers.

Finding 2.5 Antivirus Tools

An additional security tool present on John's device is a third-party antivirus product, installed under his Windows profile.

The antivirus artefacts primarily serve as contextual and timeline evidence, thereby showing that some security software was installed, but there is no indication that it successfully detected or prevented the use of the Quasar toolkit, the encrypted secret container, or the steganographic carrier BNE.png. This supports the wider finding that John's concealment strategy, which combines encryption, steganography and removable media, operated effectively beneath the level of protection provided by standard antivirus controls.

File Name	Bav.exe
Analysis	Baidu Antivirus
File Path	/img_Narcos-2.001/vol_vol7/Program Files (x86)/Baidu Security/Baidu Antivirus/5.4.3.147185.0/Bav.exe
File Size	1710576
Created Time	2019-01-29 04:18:24 SGT
Accessed Time	2019-02-02 11:03:09 SGT
Modified Time	2019-01-29 04:18:24 SGT
MD5 Hash	5c447fc7047cc905c021a2486f00c147

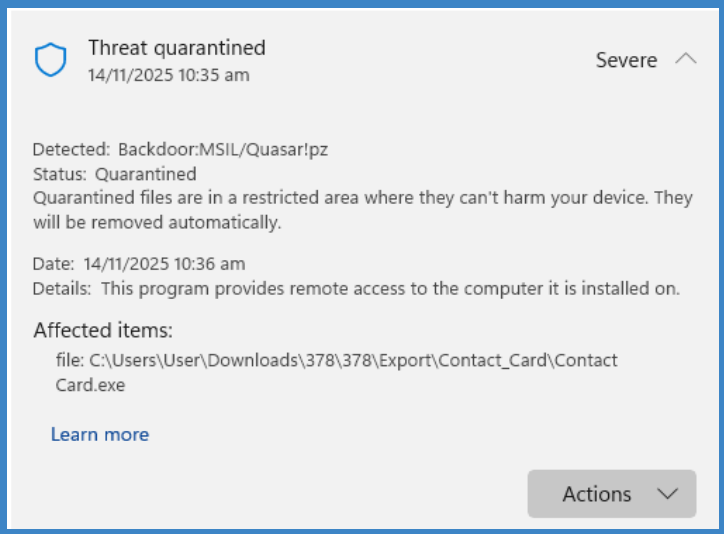
Finding 2.6 Spyware Deployment Attempt

Evidence from John's device shows that a Quasar remote-access Trojan (RAT) client was delivered to him in the form of a disguised "contact card" executable and that this spyware was used by Jane to monitor his activity. On Narcos-2 (John), the archive Contact_Card.zip was recovered from C:\Users\JohnF\Downloads\ and is known from other artefacts to have been supplied by Jane via Discord. When this archive was exported to an isolated examiner workstation and unpacked, the contained file Contact Card.exe was subjected to an on-demand scan using Microsoft Defender with current signatures. Defender detected the file as Backdoor:MSIL/Quasar!pz with a "Severe" rating and quarantined it, describing the program as providing remote access to the computer it is installed on. This confirms that the "contact card" executable on John's machine is not a benign utility but a Quasar RAT client intentionally disguised as a contact-card application.

Quasar's purpose is to give a remote operator covert control of an infected host, including keylogging, screen capture, file transfer and command execution. These capabilities match the surveillance data recovered from Narcos-3 (Jane's laptop), where Quasar 1.3.0.0 is present under C:\Users\JaneE\Downloads\Quasar v1.3.0.0\ and shellbag artefacts confirm the folder was

opened. Jane's device stores extensive keylogging-style logs that mirror John's Discord drafts, drug-related web searches, ProtonMail credentials and attempts to decrypt the TrueCrypt container secret. The tight temporal alignment between John's activity on Narcos-2 and the entries in Jane's logs, together with the confirmed Quasar client Contact Card.exe, shows that the spyware deployment was successful and that John's endpoint was under active remote surveillance operated from Jane's machine.

Targeted searches of Autopsy's Installed Programs, Run Programs, Web Downloads and Shellbag artefacts for Narcos-1 (Steve) and Narcos-2 (John) did not identify Quasar as an installed application, nor did they reveal additional Quasar-named components on Steve's image. This suggests that the Quasar client on John's machine was delivered as a stand-alone executable within Contact_Card.zip and run by John without leaving a conventional installation footprint, and that there is no direct evidence in the provided artefacts that Steve's system was itself infected with Quasar. Instead, Jane's visibility into interactions involving Steve appears to have been obtained indirectly via Quasar's monitoring of John's compromised computer. From a technical standpoint, the deployment and use of the Quasar client on John's device is indistinguishable from a criminal spyware operation; in this case, scenario information indicates that it was conducted by an undercover officer for investigative purposes.



File Name	Contact_Card.zip
File Path	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Contact_Card.zip
File Size	198604
Created Time	2019-01-30 06:01:31 SGT
Accessed Time	2019-01-30 06:03:04 SGT
Modified Time	2019-01-30 06:01:31 SGT
MD5 Hash	b4526c58d6a1a6d6bd654474a1ce8823

Finding 2.7 Summary

The examination of John Fredricksen's laptop (Narcos-2.001) shows that his device functioned as a central operational platform for the narcotics distribution network and, ultimately, as the compromised endpoint through which undercover officer Jane Esteban obtained visibility into that network. Communications artefacts demonstrate that John relied on both Discord and ProtonMail as covert-friendly channels: Discord LevelDB storage and browser history show use of Discord to receive operational files such as Contact_Card.zip, while ProtonMail history and Zone.Identifier data link his Attachments-Important, crucial to our method.zip download directly to an encrypted ProtonMail blob. These channels provided him with the "secret" file, the contact-card lure, and other material used in the operation.

Operational documents on the device show structured planning rather than casual involvement. The clients.ods spreadsheet in John's Business folder records a set of "clients" with locations, drug slang product names, regular shipment quantities and delivery frequencies, with Jane listed as a low-volume "on demand" entry. The accompanying shipping.PNG and the Steve K.PNG flight itinerary (hash-identical to flightbookings.PNG on Steve's machine) indicate that John actively consulted shipping and travel information while maintaining this ledger, tying him to the logistics of the Brisbane–Wellington run. Together, these artefacts position John as a planner and coordinator, not merely a consumer or courier.

John's handling of concealed files reveals deliberate use of layered obfuscation techniques. He received Attachments-Important, crucial to our method.zip, which contained the TrueCrypt container secret. TrueCrypt on his system, using the passphrase "ilovediving", decrypts secret to reveal Memo Things.odt ("Crystal Method"), a memo that details a 1 kg "crystal" shipment from Brisbane to Wellington, discusses transport options and risk, and references future larger orders. In parallel, he used Image Steganography with the passphrase "Elchapo2" to embed package.jpg (drug packaging imagery) inside BNE.png, a bloated derivative of 1540752698-brisbane.jpg. Additional sensitive images such as Janes Kids.jpg and package.jpg appear in VMware drag-and-drop paths, indicating active movement between host and virtual environments and providing potential leverage over Jane. These elements show John using encryption, steganography and virtualisation to protect and control operational intelligence.

Finally, spyware analysis confirms that John's endpoint was itself compromised by a Quasar remote-access Trojan under Jane's control. Jane's device holds the Quasar 1.3.0.0 toolkit and extensive keylogging logs matching John's activity. John's Contact_Card.zip contains Contact Card.exe, which was confirmed by Microsoft Defender on an examiner workstation as Backdoor:MSIL/Quasar!pz, a Quasar RAT client. This establishes that John executed a disguised Quasar client, granting Jane covert access to his machine. There is no equivalent Quasar artefact on Steve's image; Jane's insight into Steve's involvement appears to derive from monitoring John's compromised endpoint rather than directly infecting Steve. Overall, the artefacts on

Narcos-2.001 demonstrate that John was a key organiser who maintained drug client records, logistics plans and concealed operational documents, and whose own device became the primary vantage point from which the undercover officer was able to observe and record the network's activities.

Finding 3 - Analysis of Jane's Device (Narcos-3.001)

Windows artefacts from Narcos-3 (host JELAPTOP) were examined using Autopsy's Windows modules to understand the activity of the account janee. Operating System Information and OS Accounts show that janee is a normal user account created on 29 January 2019 at 03:20:32 SGT, and is the primary interactive profile on this system. Scenario documentation indicates that this user is an undercover law-enforcement officer embedded with the suspects. The analysis below therefore focuses on what the artefacts show technically, while recognising that intent (malicious vs investigative) is established by case context rather than the disk evidence alone.

Windows Web History and Web Downloads artefacts link janee to the creation of "contact card" graphics that are later seen in other parts of the investigation. Edge history records visits to Bing Images on 28 January 2019 around 21:22 SGT, including a result that references a contact-card style icon hosted on a third-party site. Immediately afterwards, Edge accesses file:///C:/Users/JaneE/Documents/contact-card-512.png, confirming that a local PNG version of the contact card was stored in Jane's Documents folder and viewed in the browser. Web Downloads and associated Zone.Identifier data show that contact-card-512.png and subsequent .ico variants (contact_card_512_CA3_icon.ico under Downloads and contact_card.ico under Pictures) were created via icoconvert.com. These artefacts collectively demonstrate that janee sourced a contact-card image from the internet, converted it into Windows icon format using an online converter, and saved those icons into her profile. From a purely forensic standpoint, this behaviour is consistent with preparing a benign icon for a contact card application, preparing a lure file, or creating visual resources for an investigative operation; the disk artefacts alone do not distinguish between those uses.

Windows artefacts also connect janee to the Quasar 1.3.0.0 remote administration toolkit. Shell Bag records from UsrClass.dat show that Explorer opened the Edge Temporary State download folder containing Quasar.v1.3.0.0.zip and subsequently browsed into an extracted Quasar v1.3.0.0 directory and its Profile subfolder. Date Created and Last Write timestamps around 28 January 2019 21:06–21:52 SGT, with additional activity on 30 January 2019, indicate that this folder tree was inspected on multiple occasions. Windows Web Downloads artefacts show multiple Quasar-related binaries (client.bin, Quasar.exe, Mono.Cecil.dll, Mono.Nat.dll, Vestris.ResourceLib.dll) stored under /Users/JaneE/Downloads/Quasar v1.3.0.0/ and again within $Recycle.Bin\…\$RN… paths, indicating that parts of the Quasar directory were later deleted or moved to the Recycle Bin. Technically, these artefacts corroborate that janee

downloaded, extracted, and explored a working copy of Quasar, and then attempted to clean up traces of that toolkit. In an undercover context, this is consistent with building, testing, or analysing the same tools used by suspects; the artefacts alone are equally compatible with a co-conspirator staging a RAT for deployment.

Additional Windows artefacts corroborate Jane's communication and storage patterns. Web History shows repeated access to https://protonmail.com/, its login page, and ProtonMail inbox URLs between late 28 January and early 29 January 2019, together with visits to login.live.com and live.com, indicating use of both ProtonMail and a Microsoft account from the JaneE profile. Web Downloads record further Internet-zone-marked files saved into her Documents and Pictures folders (including a PDF and several images), demonstrating general document and image handling around the same period as the Quasar and contact-card activity. USB Device Attached artefacts show several removable devices connected to JELAPTOP: a Seagate Backup Plus Slim portable drive on 29 January 2019, a Samsung Galaxy A5 smartphone in MTP mode on 1 February 2019, a Seagate Expansion Portable drive on 2 February 2019, and a Western Digital "Elements Portable (WDBUZG)" drive with Device ID 57584D3145373444574D314E at 2 February 2019 11:02:28 SGT. The same Western Digital device identifier appears on Steve's (Narcos-1) and John's (Narcos-2) systems, indicating that a single physical USB drive was shared across all three machines. From the perspective of corroborative Windows artefacts, this supports the interpretation that janee was fully integrated into the same toolchain and media workflow as the suspects, handling the contact-card graphics, exploring the Quasar toolkit, using ProtonMail, and sharing a removable drive that also appeared on the other Narcos systems. The undercover nature of her role is therefore a matter of external case knowledge; on the disk, her activity is intentionally indistinguishable from that of a cooperating member of the group.

Finding 3.1 Communications

A Discord log file named **000003.log** was recovered from Jane Esteban's device in the following directory: "/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Roaming/Discord/Local Storage/leveldb/000003.log"

File Name	000003.log
Analysis	Discord messages
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Roaming/Discord/Local Storage/leveldb/000003.log
File Size	17677
Created Time	2019-01-29 08:02:52 SGT
Accessed Time	2019-01-29 08:02:52 SGT
Modified Time	2019-02-02 11:07:55 SGT
MD5 Hash	396ee131e3661241c19e2d043608dbf8

The communication thread in this log reflects Jane's involvement in an undercover character. The discovered messages show an attempt to impersonate a methamphetamine buyer and develop trust with a suspected narcotics distributor. The other party is not expressly identified in the log. Nonetheless, the inclusion of the message "Yes John got it" indicates that the counterpart was intimately affiliated with John Fredricksen or was speaking indirectly via him.

Jane initiated contact by requesting "ice," in line with the drug-seeking behaviour expected of her undercover character. She reinforces this impression by saying she has "friends who are interested" and offering a "contact card."She demands "the usual," but later communications show that she does not completely grasp the situation. This lack of knowledge is consistent with her status as an undercover investigator rather than a buyer.

Jane challenges the opposing party, indicating that she does not understand what they are referring to and that the talk "sounds risky." This shows that she is attempting to obtain information or assess the counterpart's credibility.When the topic becomes ambiguous or potentially heated, Jane says she's "not willing to continue business." This suggests a tactical decision to limit exposure or disengage if her persona appears compromised.

The debate quickly intensifies. Jane's messages became more urgent or "desperate," implying that the counterpart may have challenged her undercover character or questioned her credibility. This behaviour is compatible with the reactive persona maintenance strategies utilised in undercover operations.

This communication implies that the counterpart mentioned John—possibly as a mediator or confirmation channel—and Jane's response confirms this. This establishes a link between her discussion and previous activity with John and Steve. Jane then gives her entire name and birthdate. This could be an attempt at trust-building, intentional deceit, or a tactical need in reaction to pressure from the suspected dealer. The final viewable fragment has the message "I'll be there," implying that Jane consented to a physical meeting location, albeit no metadata or map references accompany this section of the log.

The conversation is consistent with Jane's role as an undercover AFP officer investigating a potential drug distribution network. The sequence of messages, from inquiry, uncertainty, disengagement, pressure, identity disclosure, and eventual agreement, reflects real-world behaviours associated with covert persona maintenance. The letter addressing John gains credibility to the argument that Jane's undercover encounters were closely linked to the larger communication chain between John and Steve. No retrieved messages show personal engagement in criminal behaviour, rather, all action is consistent with an officer attempting to gather intelligence while preserving operational cover.

These Discord logs provide supporting documentation for Jane's undercover activities and controlled, investigative conversations with suspects. Instead of expressing a desire to participate

in drug trafficking, the discussions show information gathering activities. This evidence, which is exculpatory in terms of criminal behaviour, supports the assumption that Jane's device contains elements associated with law enforcement operations rather than illicit behaviour.

Finding 3.2 Undercover Materials

Multiple files retrieved on Jane Esteban's device suggest evidence of preparation and training, which is consistent with her acknowledged job as an undercover Australian Federal Police officer. These resources contain potential job descriptions, survival guides, operational guidance for plain-clothes officers, and AFP images. The timestamps on these files reveal consistent access and change throughout the same time period that Jane was talking with suspects, indicating their use in ongoing undercover operations.

File Name	Becoming_an_Undercover_Cop_Job_Description_Salary_Information[1].htm
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/Becoming_an_Undercover_Cop_Job_Description_Salary_Information[1].htm
File Size	113354
Created Time	2019-02-01 07:58:40 SGT
Accessed Time	2019-02-01 07:58:40 SGT
Modified Time	2019-02-01 07:58:41 SGT
MD5 Hash	a3fe407bc777952b389eecb68f17b874

This HTML page contains detailed descriptions of undercover police roles, essential abilities, risk management considerations, and salary data. Its existence shows that Jane created her operational identity by researching or reading formal job criteria. The evidence supports the theory that she was actively preparing and confirming her identity before dealing with suspects, which is consistent with her undercover AFP duties.

File Name	plainclothes-officer-survival-how-to-stay-alive-without-really-trying[1].htm
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/plainclothes-officer-survival-how-to-stay-alive-without-really-trying[1].htm
File Size	103751
Created Time	2019-02-01 08:01:11 SGT
Accessed Time	2019-02-01 08:01:11 SGT
Modified Time	2019-02-01 08:01:11 SGT
MD5 Hash	2c2cfed4aff6cd766ca797995da27afc

This file contains operational regulations for plainclothes police officers, including safety tactics, situational awareness approaches, and covert mission behavioural requirements. The fact that this file exists and is undamaged demonstrates Jane's attempts to keep herself operational. Based on its

location in the OrphanFile folder, the document could have been deleted during normal system usage or cleanup routines.

File Name	course-undercover-survival.pdf
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Documents/course-undercover-survival.pdf
File Size	140038
Created Time	2019-02-01 08:04:19 SGT
Accessed Time	2019-02-01 08:04:19 SGT
Modified Time	2019-02-01 08:04:19 SGT
MD5 Hash	fa714f0aad66bf3953d061adf2457f86

This document depicts a PDF pamphlet describing an undercover survival course, which covers training, threat awareness, and covert operations protocols. This information is certainly relevant to Jane's professional tasks, and it reinforces the assumption of her undercover assignment.

File Name	afp.png
Analysis	Australian Federal Police Badge
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/afp.png
File Size	68697 bytes
Created Time	2019-01-29 06:46:06 SGT
Accessed Time	2019-01-29 06:46:06 SGT
Modified Time	2019-01-29 06:47:48 SGT
MD5 Hash	95ef21d225d0902bc8cbd64fc113740a

File Name	background.jpg
Analysis	Australian Federal Police Logo
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/background.jpg
File Size	27438 bytes
Created Time	2019-01-29 06:47:33 SGT
Accessed Time	2019-01-29 06:47:33 SGT
Modified Time	2019-01-29 06:47:33 SGT
MD5 Hash	e13f7de15d62717449376d6a54ce3d5e

The images illustrate AFP's emblem and logo. The AFP-related photographs reinforce Jane's assumed verified law enforcement status. The existence of these images in Jane's device, although not concrete, gives more credibility towards her claim of being an undercover police officer. It also reveals that the device potentially belongs to an active-duty officer who retained materials relevant to her investigation position.

The files combined demonstrate a thorough research and operational planning required to maintain a credible undercover identity. The timestamps correspond perfectly to the operative period (late January to early February 2019), when Jane was talking with suspects as part of her assignment. The presence of several files in $OrphanFiles shows that they were deleted during a possible

routine activity or browser cleanup, which is compatible with high-risk covert activities rather than the concealing of unlawful behaviour.

These materials are exculpatory. They affirm, to an extent, that Jane's device contains legitimate law enforcement training material and operational references, which support her employment as an undercover AFP officer. None of the files indicate criminal intent. Instead, they show the preparation, study, and awareness needed for clandestine penetration of a narcotics trafficking network.

Finding 3.3 Surveillance and Keylogging

Between January 30, 2019, and February 2, 2019, a number of consecutive days' worth of keylogging artefacts were found on Jane Esteban's device. These logs, which came from John Fredricksen's computer, include thorough recordings of keyboard input, window activity, browser interactions, and program use. Several logs were discovered in both active system regions and within $OrphanFiles, suggesting that they may have been deleted during routine system cleanup. The logs are compatible with data produced by popular remote keylogging or remote administration tools like Quasar.

The presence of the file contact_card.zip on John's device (see Finding 2.4) and concurrent surveillance logs on Jane's system indicate that the keylogger was presumably delivered through this file, consistent with covert monitoring techniques used in undercover operations, even though the exact deployment mechanism was not recovered.

Finding 3.3.1 Recovered Keylogging Logs

Several log files have been discovered, including several.txt files (such as f[2].txt, f[3].txt, f[4].txt, and f[5].txt) and daily HTML-formatted logs (such as 01-30-2019.html, 01-31-2019.html, 02-01-2019.html, and 02-02-2019.html). These files consists of:

- Keypress events (Backspace, Enter, Left and Right Arrows, and Alphanumeric Inputs)
- Window Focus Changes
- Browser Search Queries
- Email and Login Activities
- Password Entries
- Discord Message Composition and Edits

The consistency of timestamps and user behaviour strongly suggests that John's system was the sole source of these entries.

Finding 3.3.2 Content Observed within Logs

The logs revealed multiple instances of John entering or attempting to input credentials, including:

- ProtonMail login: heresjohnny1@protonmail.com
- Password entry: John1234
- Multiple attempts to decrypt the file “secret” using various passwords.

These occurrences reveal active monitoring of John's authentication attempts and access to his encrypted data.

The records documented John's keystrokes as he typed Discord messages, including:

- Message drafting
- Edits and deletions
- Aggressive or threatening language
- Operational discussions
- Communications aligned with activities observed on Steve’s device

These fragments replicate the Discord logs discovered earlier in the inquiry and provide insight into John's real-time communication behaviour. Captured browser searches included:

- Drug concealment techniques
- Methamphetamine terminology
- Logistics and supply-related queries
- Attempts to locate or research criminal content online

These entries highlight John's role in narcotics management and strategy.

Finding 3.3.3 File Origins

Examination of the recovered log files on Narcos-3 shows that many of them have a structure consistent with keyboard-monitoring modules used by tools such as Quasar. The logs are stored in timestamped HTML files containing ordered key-event records and consistent keystroke capture patterns, rather than simple text notes or browser artefacts. The content includes typed URLs, Discord message drafts, ProtonMail credentials and command-line fragments that match activity known to have taken place on John’s device.

Additional log files were identified under Autopsy’s $OrphanFiles view, indicating that their original directory entries have been removed or overwritten. This is consistent with keylogger outputs that were actively written to disk and later deleted or moved, for example during manual

tidy-up or when browser/system cleanup tools altered the directory structure. Despite this, the orphaned data blocks remain recoverable and preserve both timestamps and content.

The combination of log structure, content and timeframe indicates that John's computer was being monitored remotely and that the resulting surveillance data was stored on Jane's device. Scenario documentation identifies Jane as an undercover AFP officer, and in that context the logs are best interpreted as outputs from a controlled surveillance operation rather than as evidence that Jane herself was acting maliciously. The same logs are directly incriminating for John: they capture unfiltered user behaviour including drug-related web activity, credential use, attempts to work with the encrypted secret file and communications planning that tie back to artefacts found on Steve's machine.

Taken together, these artefacts are exculpatory for Jane, which supports her role as an investigator conducting covert monitoring, and inculpatory for John, as they document his real-time involvement in drug planning, communication and concealment. The extensive, timestamped keylogging material also demonstrates the sophistication of the surveillance in this case by providing a detailed, chronological view of John's digital behaviour and state of mind throughout the investigative period.

File Name	01-30-2019.html
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-30-2019.html
File Size	2777
Created Time	2019-01-30 09:41:16 SGT
Accessed Time	2019-01-30 09:41:16 SGT
Modified Time	2019-02-02 10:52:34 SGT
MD5 Hash	fd3a044bee4d5af72faba9630867d4fd

File Name	01-31-2019.html
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-31-2019.html
File Size	6486
Created Time	2019-02-02 10:52:34 SGT
Accessed Time	2019-02-02 10:52:34 SGT
Modified Time	2019-02-02 10:52:34 SGT
MD5 Hash	948b4c559b944d6c0f2e1da8e285d835

File Name	02-01-2019.html
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-01-2019.html
File Size	5733
Created Time	2019-02-02 10:52:34 SGT
Accessed Time	2019-02-02 10:52:34 SGT
Modified Time	2019-02-02 10:52:34 SGT
MD5 Hash	f51aa5754e54915e93d329a3eefdc042

File Name	02-02-2019.html
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-02-2019.html
File Size	5014
Created Time	2019-02-02 10:52:34 SGT
Accessed Time	2019-02-02 10:52:34 SGT
Modified Time	2019-02-02 10:52:34 SGT
MD5 Hash	7edea768c94997ed685bf7bc51cc325a

Finding 3.4 Spyware Propagator

The malware identified in this case is the Quasar remote administration toolkit (RAT), used by undercover officer Jane Esteban to conduct covert surveillance against suspect John Fredricksen. On Narcos-3 (Jane's device), a full Quasar 1.3.0.0 toolkit is present under C:\Users\JaneE\Downloads\Quasar v1.3.0.0\, including Quasar.exe, client.bin and associated .NET libraries. Shellbag artefacts confirm that this folder was opened in Explorer, demonstrating that Quasar was actively handled on Jane's machine rather than being a dormant download. On Narcos-2 (John's device), the archive Contact_Card.zip was recovered from C:\Users\JohnF\Downloads\ and is known to have been supplied by Jane via Discord. When the embedded Contact Card.exe was exported to an isolated examiner workstation and scanned with Microsoft Defender (current signatures), it was detected and quarantined as Backdoor:MSIL/Quasar!pz, with the product description stating that the program "provides remote access to the computer it is installed on." This confirms that Contact Card.exe is a Quasar RAT client disguised as a benign "contact card" executable.

Quasar is a .NET-based remote administration Trojan which, once a client stub is executed on a target, allows the controller to perform keylogging, screen capture, file and process manipulation, credential theft and arbitrary command execution over an encrypted channel. These capabilities match the keylogging-style logs recovered from Jane's device: the logs record John's Discord drafts, drug-related web searches, ProtonMail credentials and attempts to decrypt

the secret TrueCrypt container. The combination of (i) the Quasar builder and controller on Jane's device, (ii) the confirmed Quasar client Contact Card.exe derived from John's Contact_Card.zip, and (iii) detailed records of John's activity on Jane's machine, demonstrates that Quasar was used as spyware to monitor John's endpoint throughout the relevant period. Interactions between John and Steve are visible to Jane not because Steve's machine was demonstrably infected, but because John's compromised computer captured the communications in which Steve participated.

Targeted searches of Autopsy's Installed Programs, Run Programs, Web Downloads and Shellbag artefacts for Narcos-1 (Steve) and Narcos-2 (John) found no evidence of Quasar being installed as a conventional application and no additional Quasar-named components on Steve's image. This suggests that Quasar was deployed as a stand-alone client delivered to John as Contact Card.exe, rather than as a globally installed program, and that Steve's device was not proven to be running a Quasar client in the material available. However, the removable-media timeline shows that external drives (a Seagate disk shared between Jane and Steve, and a Western Digital Elements drive shared between Steve and John) circulated between the same hosts during the period when Quasar was acquired and used. These devices provide a feasible secondary propagation route for additional Quasar clients or related tooling, even though no such client was recovered from Steve's disk image.

The vulnerability that allowed Quasar to function in this environment is primarily human and procedural rather than a software flaw. John executed an untrusted executable (Contact Card.exe) received in a ZIP archive over Discord, relying on appearance and context rather than provenance; there is no indication of application whitelisting or execution control to prevent this. Conventional antivirus was present but did not prevent Quasar's deployment at the time of infection, and would not be able to inspect encrypted data stored inside TrueCrypt containers or steganographically hidden in images. The use of encrypted channels (ProtonMail, Discord) and removable media further bypassed perimeter security controls. In effect, Quasar exploited a security posture where users were free to run arbitrary code from archives and USB drives and where endpoint monitoring and application control were insufficient to detect or block a disguised remote-access backdoor.

Finding 3.5 Online Research

The forensic investigation of Jane Esteban's device found numerous files and webpages relating to drug trade knowledge, methamphetamine culture, and covert enforcement operations. These materials are appropriate for the research and persona-development needs of an undercover AFP officer assigned to infiltrate narcotics distribution networks. The collected items include news articles, forum pages, drug lessons, and operational research documents. Several were found in the $OrphanFiles directory, indicating that they had been accessed and then erased via ordinary system activity or cleanup tools.

File Name	drug-dealing-men[1].jpg
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/drug-dealing-men[1].jpg
File Size	105156
Created Time	2019-02-01 07:06:44 SGT
Accessed Time	2019-02-01 07:06:44 SGT
Modified Time	2019-02-01 07:06:44 SGT
MD5 Hash	2b53bfe7ab249435d9a2b9d80f473a0f

An image file displaying a newspaper article on drug dealers. The article appears to have been saved or downloaded from an online source. The essay most likely provided context for studying offender profiles, behavioural trends, and case similarities—all of which are important preparation steps in the formation of undercover roles.

Multiple .jpg, .png and .htm files document, present in Jane's device, containing:

- Methamphetamine Crystals
- Explantory Diagrams
- Meth Details and Descriptions
- Books about Methamphetamine and Crystal Meths
- Informative Slides about Methamphetamine

File Name	how-to-be-drug-dealer[1].htm
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/how-to-be-drug-dealer[1].htm
File Size	81778
Created Time	2019-02-01 07:06:40 SGT
Accessed Time	2019-02-01 07:06:40 SGT
Modified Time	2019-02-01 07:06:40 SGT
MD5 Hash	618813dfadec604c1bd741e1845848f7

Such information would be required for Jane to detect drug references, avoid revealing operational flaws while talking with suspects, and retain credibility when posing as a buyer or associate.

Jane's recovered web research strongly implies and strengthens her identity as an undercover AFP officer. Her files show her actively preparing for interactions with suppliers and drug-related

individuals, gaining a better understanding of the methamphetamine culture and offender behaviour, and becoming acquainted with covert surveillance principles and ongoing maintenance of an undercover operative.

These artefacts are not incriminating. Instead, they provide context for Jane's operational readiness and explain her ability to communicate successfully with suspects like John and Steve. They do not represent unlawful drug interest or engagement, but rather the professional activities required for undercover drug investigations. Their presence supports the broader narrative that Jane's device activity was in accordance with AFP directives and intelligence collection specifications.

Finding 3.6 Summary

The forensic examination of Jane Esteban's device (Narcos-3.001) identified a cohesive set of artefacts that align with scenario information describing her as an undercover Australian Federal Police (AFP) officer embedded in a narcotics distribution network. The artefacts show persona building, operational preparation, covert communication and surveillance activity focused on a key suspect, John Fredricksen. While several tools and techniques identified on the device (e.g. remote administration/keylogging software) would normally be classed as malicious, in this case their use is consistent with law-enforcement surveillance rather than personal criminal gain.

Discord artefacts on Narcos-3 show Jane engaging in controlled undercover contact, presenting herself as a potential methamphetamine buyer to gain credibility with a suspected dealer. The message sequence, which includes initial enquiry, hesitation, withdrawal, pressure from the dealer and eventual agreement; is consistent with maintaining an undercover persona under threat. References to John in this conversation (for example, "Yes, John got it") situate her communications within the same distribution network evidenced on the other suspects' devices.

The device also contained multiple research and operational documents, including undercover survival manuals, plain-clothes policing tactics, and AFP-related images. These materials are consistent with formal law-enforcement training and preparation for covert operations, and help explain the knowledge and terminology Jane applies when interacting with suspects. Scenario documentation explicitly identifies Jane as an AFP officer, and the presence of these resources on her device corroborates that role.

In addition, the device holds extensive keylogging-style logs of activity attributable to John's computer, including Discord draft messages, drug-related web searches, ProtonMail credentials and attempts to access or interpret the "secret" file. The volume, timestamps and content of these logs indicate systematic monitoring rather than opportunistic theft. Given the presence of a Quasar 1.3.0.0 remote administration toolkit on Jane's device, a technically plausible explanation is that a RAT/keylogger client deployed from a controlled payload (potentially linked to *contact_card.zip*) captured John's activity and transmitted it to Jane's machine. The precise deployment vector cannot be conclusively proven from the available artefacts, but the

combination of Quasar, the logs, and the known relationship between Jane and John strongly supports an interpretation of controlled surveillance.

No files were identified on Jane's device that could not be reasonably explained as either personal artefacts or tools and outputs related to an undercover operation. Overall, the Windows artefacts, communication records and surveillance logs portray Jane's digital activity as structured information gathering, planning and monitoring focused on individuals in the narcotics network. When considered alongside case documentation naming her as AFP, these findings are consistent with a law-enforcement investigative role and support an exculpatory interpretation of her conduct.

Summary and Conclusion

This investigation examined six digital evidence sources acquired from three Windows systems associated with the case codenamed Narcos: Steve Kowhai's desktop, John Fredricksen's laptop, and Jane Esteban's laptop. Using Autopsy 4.22.1, Magnet AXIOM, FTK Imager and Windows hashing tools, the team reconstructed user activity, correlated artefacts across hosts and timelines, and attributed behaviour to specific user accounts. Despite several processing issues and partially corrupted or tampering of the files provided (noting the table below showing mismatched hash files of the original files and the ones we were give), a coherent picture emerged of a transnational methamphetamine distribution operation centered on John and Steve, with Jane acting as an embedded AFP undercover operative rather than a co-conspirator.

Actor Name	File Name	MD5 Hash Value provided in the Assignment Brief	MD5 Hash Value Computed from Supplied File	Match
Steve Kowhai	Narcos-1a.001-021	996182c381ec9e7025f40519107615e4	063a8227da15f7e992e011ce62547f3b	No
	Narcos-Mem-1a.001-003	3469089f9a26b0b51a4ee985cd1c3008	fd66ca5022d5e4f645161947b47db54e	No
Jane Estaban	Narcos-3a.001-021	ce707bf783dde13ed42196cd6e473083	dc8ae21d07283176f2c6748e2fb1ab5b	No
	Narcos-Mem-3a.001-003	7c155dae658fb059586b3ab5144e21d2	096a0c0461a7e36cd6b245c72475d17a	No
John Fredricksen	Narcos-2a.001-021	56823dee9b24a40407bec184f80261c2	f95101fa223c6bedce0edda9c72b0054	No
	Narcos-Mem-2a.001-003	fd7027c7bffedd653226355467c2b1ef	5dda265b1730edaac9d03aac6608075a	No

On Steve's desktop, the file system artefacts, Windows outputs, and recovered images collectively showed that he was operating as a downstream distributor and logistics node in Wellington. Images depicting crystal meth, drug money, route diagrams, and a library drop-off location, combined with a flight booking from Brisbane to Wellington, closely matched the physical seizure of narcotics and intelligence on parcel movements. Steve had installed and executed TrueCrypt, Image Steganography and CCleaner during the critical period, while also

connecting multiple USB storage devices. These artefacts support the conclusion that he encrypted, concealed, and attempted to wipe incriminating material (including the file “secret”) and was suspected to have used removable media as a transport channel for both operational documents and concealed images.

John’s laptop contained the strategic layer of the operation. A client.ods spreadsheet listed multiple international clients, their locations, product codes and regular shipment volumes, providing a structured view of a wholesale-level supply network rather than casual personal use. Business-related images and documents (such as shipping.png and Steve K.png) linked those abstract entries to concrete shipments and travel by Steve. John also prepared and used steganography and encryption; installing Image Steganography and TrueCrypt, he created BNE.png from a benign Brisbane photograph, and passed both BNE.png and a TrueCrypt container to Steve through shared media and zip archives. These choices demonstrate deliberate tradecraft aimed at hiding operational instructions and sensitive content from casual inspection.

Analysis of Jane’s system, by contrast, reveals a different pattern, as her device held undercover training material, AFP branding, and research on plain-clothes work and drug dealing methods, consistent with persona construction and operational planning by law enforcement. Most significantly, Jane’s Downloads folder contained Quasar v1.3.0.0, configured with a client entry “JohnF@JOHNFLAPTOP1_4DD90B0” and a series of HTML keylogging logs dated 30 January to 2 February 2019. These logs captured John’s keystrokes, draft Discord messages, drug-related searches, credential use and attempts to work the “secret” file. In combination with the contact_card.zip artefact present on both Jane’s and John’s systems, and Baidu Antivirus activity on John’s host, the evidence supports that John’s laptop was infected with a Quasar RAT payload, controlled from Jane’s machine to support lawful covert surveillance. No equivalent Quasar installation or telemetry path was identified on Steve’s machine.

Cross-host correlation of Windows artefacts and device timelines reinforced these role assessments. On Steve’s desktop, OS Accounts, Run Programs, Web History, Web Downloads, Recent Documents, USB Device Attached and Discord leveldb record converged on a short but intensive period between 29 January and 2 February 2019, during which Steve created and accessed secret, executed TrueCrypt, Image Steganography and CCleaner, and attached multiple USB disks. On John’s laptop similar artefacts placed his use of clients.ods, steganographic tolling, BNE.png and zipped attachments within that same window, aligning with Discord exchanges and with the appearance of the BNE.png steganographic carrier and TrueCrypt secret container on Steve’s host. On Jane’s laptop, the Quasar logs, undercover documentation and Discord conversations align temporally with John’s preparation, communication with Steve, and attempts to handle the secret container. Together, these Windows artefacts and timelines demonstrate a coordinated chain: John prepares and orchestrates, Steve executes logistics and clean-up, and Jan monitors and infiltrates as a law-enforcement asset.

There are limitations. Some file system entries, such as Steve's secret, are now zero-byte placeholders with wiped timestamps, indicating intentional removal beyond straightforward recovery. Some Discord content could only be partially reconstructed from leveldb logs rather than fully formatted messages. Certain Quasar-related actions on John's host are inferred from logs stored on Jane's machine and from contact_card.zip rather than from an intact Quasar client binary on the Narcos-2 image. These gaps do not undermine the overall narrative but mean that some chains of events are supported by converging circumstantial digital artefacts rather than a single definitive object.

Overall, the digital evidence strongly supports the following conclusions. Steve Kowhai acted as an active participant in the methamphetamine distribution network, handling shipments, preparing drop-offs, and deliberatively using encryption, steganography and system cleaning to conceal his role. John Fredricksen functioned as the primary organiser and upstream coordinator, maintaining a structured client list, issuing instructions, preparing steganographic carriers and encrypted containers, and coordinating with Steve to move product and cash. Jane Esteban, while deeply embedded in the same communication and file-exchange environment, is best characterised as an undercover AFP officer: her device hosts law-enforcement training material, Quasar command-and-control infrastructure and keylogging logs that incriminate John, rather than artefacts showing personal involvement in drug trafficking. The evidence collected here is internally consistent across hosts and time, aligns with physical seizures and intelligence, and provides a robust digital foundation to support prosecution of John and Steve while exonerating Jane in her professional capacity as an undercover officer.

Appendix

Timeline Progress

Date	Task	Duration
14-Oct-2025	Ingestion of Narcos-1.001	6 hours
15-Oct-2025	Ingestion of Narcos-2.001	6 hours
16-Oct-2025	Ingestion of Narcos-3.001	8 hours
20-Oct-2025	Ingestion of Steve Primary (Day 1)	45 mins
20-Oct-2025	Ingestion of Steve Primary (Day 2)	40 mins
20-Oct-2025	Ingestion of Steve Primary (Day 4)	40 mins
20-Oct-2025	Ingestion of Steve Alt (Day 1)	40 mins
20-Oct-2025	Ingestion of Steve Alt (Day 2)	40 mins
20-Oct-2025	Ingestion of Steve Alt (Day 3)	35 mins
20-Oct-2025	Ingestion of Steve Alt (Day 4)	35 mins
20-Oct-2025	Ingestion of Steve Alt (Day 5)	35 mins
20-Oct-2025	Ingestion of John Primary (Day 1)	40 mins
20-Oct-2025	Ingestion of John Primary (Day 2)	40 mins
20-Oct-2025	Ingestion of John Primary (Day 3)	45 mins
20-Oct-2025	Ingestion of John Primary (Day 4)	35 mins
22-Oct-2025	Ingestion of John Alt (Day 1)	45 mins
22-Oct-2025	Ingestion of John Alt (Day 2)	45 mins
22-Oct-2025	Ingestion of Jane Primary (Day 2)	45 mins
22-Oct-2025	Ingestion of Jane Primary (Day 3)	40 mins
22-Oct-2025	Ingestion of Jane Primary (Day 4)	45 mins
22-Oct-2025	Ingestion of Jane Alt (Day 1)	40 mins
22-Oct-2025	Ingestion of Jane Alt (Day 2)	40 mins

22-Oct-2025	Ingestion of Jane Alt (Day 3)	45 mins
22-Oct-2025	Ingestion of Jane Alt (Day 4)	45 mins
25-Oct-2025	Analysis of Narcos-1.001	8 hours
26-Oct-2025	Analysis of Narcos-2.001	8 hours
28-Oct-2025	Analysis of Narcos-3.001	10 hours
30-Oct-2025	Analysis of Steve Primary Files	2 hours
30-Oct-2025	Analysis of Steve Alt Files	2.5 hours
31-Oct-2025	Analysis of John Primary Files	2 hours
31-Oct-2025	Analysis of John Alt Files	1 hours
31-Oct-2025	Analysis of Jane Primary Files	2 hours
31-Oct-2025	Analysis of Jane Alt Files	2.5 hours
05-Nov-2025	Writing of Introduction, Executive Summary and Tables of Content	1 hours
05-Nov-2025	Writing of List of Acronyms and Tools	0.5 hours
07-Nov-2025	Writing of Methodology Details	1.5 hours
09-Nov-2025	Writing of Finding 1	3 hours
11-Nov-2025	Writing of Finding 2	3.5 hours
12-Nov-2025	Writing of Finding 3	4 hours
13-Nov-2025	Summary and Conclusion	1 hours
14-Oct-2025 - 14-Nov-2025	Appendix (Timeline Progress)	-
14-Oct-2025 - 14-Nov-2025	Appendix (Tables of Findings)	-

Table of Findings

File Name	File Path/Location
000006.log	/img_Narcos-1.001/vol7/Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/
BNE.png	/img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Misc/BNE.png
secret	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/secret
Image Steganography.exe	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/CPascoe/Image Steganography/ProgData/Image Steganography.exe
price-meth-bust-4.jpg	/img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/price-meth-bust-4.jpg
620x349.jpg	/img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/620x349.jpg
$RA3IE5E.jpg	/img_Narcos-1.001/vol_vol7/$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/$RA3IE5E.jpg
$RIIK1AS.jpg	/img_Narcos-1.001/vol_vol7/$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/$RIIK1AS.jpg
airport crystals.jpg	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/airport crystals.jpg
dropoff.jpg	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/dropoff.jpg

flightbookings.PNG	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/flightbookings.PNG
Method run.jpg	/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/Method run.jpg
CCleaner.exe	/img_Narcos-1.001/vol_vol7/Program Files/CCleaner/CCleaner.exe
dialog.min[1].js	/img_Narcos-1.001/vol_vol7/$OrphanFiles/dialog.min[1].js
hqdefaultPBGJZ8A2.jpg	/img_Narcos-1.001/vol_vol7/$OrphanFiles/hqdefaultPBGJZ8A2.jpg
jquery.themepunch.revolution.min[1].js	/img_Narcos-1.001/vol_vol7/$OrphanFiles/jquery.themepunch.revolution.min[1].js
VMware	/img_Narcos-1.001/vol_vol7/Program Files/VMware
000007.ldb	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000007.ldb
000013.log	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000013.log
000014.ldb	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000014.ldb
clients.ods	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/clients.ods
shipping.PNG	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/shipping.PNG
Steve K.PNG	/img_Narcos-2.001/vol_vol7/Users/JohnF/Documents/Business/Steve K.PNG

Attachments-Important, crucial to our method.zip	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Attachments-Important, crucial to our method.zip
secret	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Attachments-Important, crucial to our method.zip/secret
Janes Kids.jpg	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Local/Temp/vmware-JohnF/VMwareDnD/2cce6695/Janes Kids.jpg
package.jpg	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Local/Temp/vmware-JohnF/VMwareDnD/770cd05f/package.jpg
BNE.png	/img_Narcos-2.001/vol_vol7/Users/JohnF/Pictures/BNE.png
1540752698-brisbane.jpg	/img_Narcos-2.001/vol_vol7/Users/JohnF/Pictures/1540752698-brisbane.jpg
Image Steganography.exe	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/CPascoe/Image Steganography/ProgData/Image Steganography.exe
TrueCrypt.exe	/img_Narcos-2.001/vol_vol7/Program Files/TrueCrypt/TrueCrypt.exe
Bav.exe	/img_Narcos-2.001/vol_vol7/Program Files (x86)/Baidu Security/Baidu Antivirus/5.4.3.147185.0/Bav.exe
Contact_Card.zip	/img_Narcos-2.001/vol_vol7/Users/JohnF/Downloads/Contact_Card.zip
E:\Memo Things.odt	/img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roami

	ng/Microsoft/Windows/Recent/AutomaticDestinations/5f7b5f1e01b83767.automaticDestinations-ms/Memo Things.odt.lnk
000003.log	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Roaming/Discord/Local Storage/leveldb/000003.log
Becoming_an_Undercover_Cop_Job_Description_Salary_Information[1].htm	/img_Narcos-3.001/vol_vol7/$OrphanFiles/Becoming_an_Undercover_Cop_Job_Description_Salary_Information[1].htm
plainclothes-officer-survival-how-to-stay-alive-without-really-trying[1].htm	/img_Narcos-3.001/vol_vol7/$OrphanFiles/plainclothes-officer-survival-how-to-stay-alive-without-really-trying[1].htm
course-undercover-survival.pdf	/img_Narcos-3.001/vol_vol7/Users/JaneE/Documents/course-undercover-survival.pdf
afp.png	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/afp.png
background.jpg	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/background.jpg
01-30-2019.html	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-30-2019.html
01-31-2019.html	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-31-2019.html
02-01-2019.html	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-01-2019.html

02-02-2019.html	/img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-02-2019.html
drug-dealing-men[1].jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/drug-dealing-men[1].jpg
how-to-be-drug-dealer[1].htm	/img_Narcos-3.001/vol_vol7/$OrphanFiles/how-to-be-drug-dealer[1].htm
cb=gapi[1].js	/img_Narcos-1.001/vol_vol7/$OrphanFiles/cb=gapi[1].js
images027LPBI1.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images027LPBI1.jpg
images0MXBBJB0.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images0MXBBJB0.jpg
images0YKJCBN1.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images0YKJCBN1.jpg
images14ZUARLM.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images14ZUARLM.jpg
images224PSMX6.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images224PSMX6.jpg
images266MJ8E5.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images266MJ8E5.jpg
images2YVU4AOW.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images2YVU4AOW.jpg
images3BSBYOUY.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images3BSBYOUY.jpg

images4KCGCDST.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images4KCGCDST.jpg
images51Y4LVYF.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images51Y4LVYF.jpg
images9LTP37YU.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images9LTP37YU.jpg
imagesIWCSJ2FS.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesIWCSJ2FS.jpg
imagesJ2DI9IW7.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesJ2DI9IW7.jpg
imagesX367TOHU.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesX367TOHU.jpg
imagesXBZOXMH0.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesXBZOXMH0.jpg
imagesY1RZAFQC.jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesY1RZAFQC.jpg
images[1].png	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images[1].png
145382519_meth-rgb_t600[1].jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/145382519_meth-rgb_t600[1].jpg
6357449449870807 57-Escobar-Meth-pic-1[1].jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/6357449449870807 57-Escobar-Meth-pic-1[1].jpg
7b74bb6d-2118-405a-9c51-8a7a5925c5e7-meth-12_1[1].jpg	/img_Narcos-3.001/vol_vol7/$OrphanFiles/7b74bb6d-2118-405a-9c51-8a7a5925c5e7-meth-12_1[1].jpg

crys1.jpg	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys1.jpg
crys2.jpg	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys2.jpg
crys3.jpg	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys3.jpg

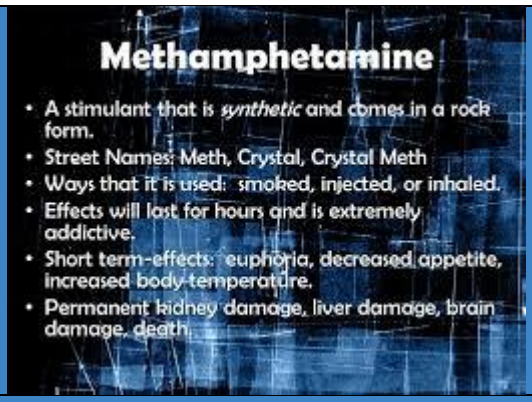
File Name	cb=gapi[1].js
Analysis	Steve's research on Wellington's Expressway
File Location	/img_Narcos-1.001/vol_vol7/$OrphanFiles/cb=gapi[1].js
File Type	.js file
File Size	150921 bytes
Created Time	2019-01-30 05:04:49 SGT
Accessed Time	2019-01-30 05:04:49 SGT
Modified Time	2019-01-30 05:04:49 SGT
MD5 Hash	dd41a9cd0d361878f946da307bbf8582

File Name	images027LPBI1.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images027LPBI1.jpg
File Size	6668
Created Time	2019-01-31 09:10:18 SGT
Accessed Time	2019-01-31 09:10:18 SGT
Modified Time	2019-01-31 09:10:18 SGT
MD5 Hash	bba634de10b820875c8a94952aa7a35e

File Name	images0MXBBJB0.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images0MXBBJB0.jpg
File Size	8589
Created Time	2019-01-31 09:11:52 SGT
Accessed Time	2019-01-31 09:11:52 SGT
Modified Time	2019-01-31 09:11:52 SGT
MD5 Hash	f2639cf4e7ff3ec7be92d10337b0f126

File Name	images0YKJCBN1.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images0YKJCBN1.jpg
File Size	4892
Created Time	2019-01-31 09:10:18 SGT
Accessed Time	2019-01-31 09:10:18 SGT
Modified Time	2019-01-31 09:10:18 SGT
MD5 Hash	d1f7bcf2e04f3828b48cd788fba4b5fb

File Name	images14ZUARLM.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images14ZUARLM.jpg
File Size	6654
Created Time	2019-01-31 09:10:19 SGT
Accessed Time	2019-01-31 09:10:19 SGT
Modified Time	2019-01-31 09:10:19 SGT
MD5 Hash	98f666faa353f7ce321f701e1cbb170d



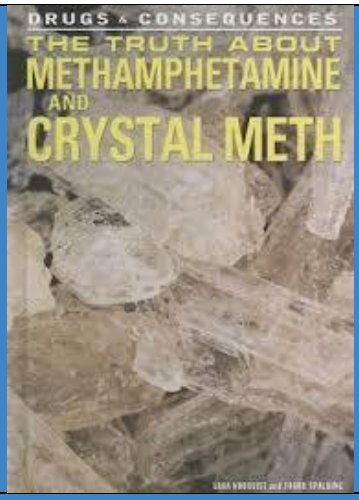
File Name	images224PSMX6.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images224PSMX6.jpg
File Size	17739
Created Time	2019-01-31 09:11:52 SGT
Accessed Time	2019-01-31 09:11:52 SGT
Modified Time	2019-01-31 09:11:52 SGT
MD5 Hash	eab22fe6e6962e00dec2af4241968938

File Name	images266MJ8E5.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images266MJ8E5.jpg
File Size	7082
Created Time	2019-01-31 09:11:59 SGT
Accessed Time	2019-01-31 09:11:59 SGT
Modified Time	2019-01-31 09:11:59 SGT
MD5 Hash	dba688a869299215f5b927cfc2c3f55c

File Name	images2YVU4AOW.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images2YVU4AOW.jpg
File Size	9908
Created Time	2019-01-31 09:11:51 SGT
Accessed Time	2019-01-31 09:11:51 SGT
Modified Time	9-01-31 09:11:52 SGT
MD5 Hash	43962be75c665825418098aea2ad6064

File Name	images3BSBYOUY.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images3BSBYOUY.jpg
File Size	8133
Created Time	2019-01-31 09:10:20 SGT
Accessed Time	2019-01-31 09:10:20 SGT
Modified Time	2019-01-31 09:10:20 SGT
MD5 Hash	62bdc9bb52fcdb1c940f2c08bf1bd0c1

File Name	images4KCGCDST.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images4KCGCDST.jpg
File Size	15466
Created Time	2019-01-31 09:10:20 SGT
Accessed Time	2019-01-31 09:10:20 SGT
Modified Time	Mo2019-01-31 09:10:20 SGT
MD5 Hash	eb19d49ea02298a238efdbf56a37acc3



File Name	images51Y4LVYF.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images51Y4LVYF.jpg
File Size	9052
Created Time	2019-01-31 09:10:18 SGT
Accessed Time	2019-01-31 09:10:18 SGT
Modified Time	2019-01-31 09:10:18 SGT
MD5 Hash	e76f878449054dcce4e5fb6cbe146b0d

File Name	images9LTP37YU.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images9LTP37YU.jpg
File Size	15347
Created Time	2019-01-31 09:11:59 SGT
Accessed Time	2019-01-31 09:11:59 SGT
Modified Time	2019-01-31 09:11:59 SGT
MD5 Hash	dc8ae5d5743de5abe32a0fb1136cdd42

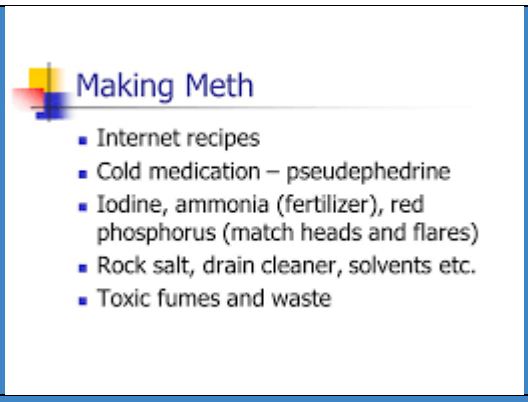
File Name	imagesIWCSJ2FS.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesIWCSJ2FS.jpg
File Size	9246
Created Time	2019-01-31 09:12:00 SGT
Accessed Time	2019-01-31 09:12:00 SGT
Modified Time	2019-01-31 09:12:00 SGT
MD5 Hash	41384db6c5c730ef3f8f127e5ab1da2b

File Name	imagesJ2DI9IW7.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesJ2DI9IW7.jpg
File Size	9906
Created Time	2019-01-31 09:11:51 SGT
Accessed Time	2019-01-31 09:11:51 SGT
Modified Time	2019-01-31 09:11:51 SGT
MD5 Hash	e237fd44f3901aa951c3de926b2b2fdc

File Name	imagesX367TOHU.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesX367TOHU.jpg
File Size	10029
Created Time	2019-01-31 09:10:18 SGT
Accessed Time	2019-01-31 09:10:18 SGT
Modified Time	2019-01-31 09:10:18 SGT
MD5 Hash	777473c663852e2767b03b2e598c8e74

File Name	imagesXBZOXMH0.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesXBZOXMH0.jpg
File Size	
Created Time	2019-01-31 09:09:13 SGT
Accessed Time	2019-01-31 09:09:13 SGT
Modified Time	2019-01-31 09:09:13 SGT
MD5 Hash	c1a2678e8ac972698bd7857924fed5bf

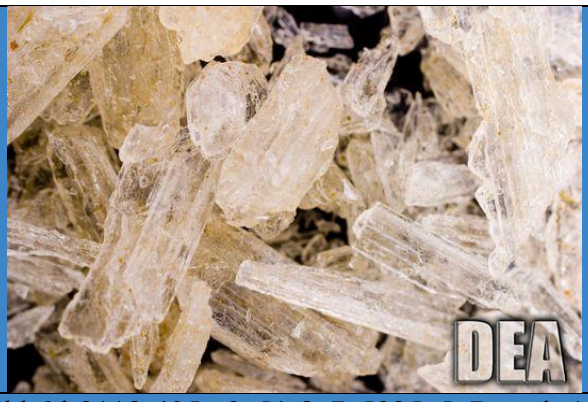
File Name	imagesY1RZAFQC.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/imagesY1RZAFQC.jpg
File Size	17603
Created Time	2019-01-31 09:11:59 SGT
Accessed Time	2019-01-31 09:11:59 SGT
Modified Time	2019-01-31 09:11:59 SGT
MD5 Hash	33fec94a65ffff7f50b6c6cd9b92207c



File Name	images[1].png
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/images[1].png
File Size	7596
Created Time	2019-01-31 09:11:59 SGT
Accessed Time	2019-01-31 09:11:59 SGT
Modified Time	2019-01-31 09:11:59 SGT
MD5 Hash	3697dcdad04c6d8749f6ed62002469c8

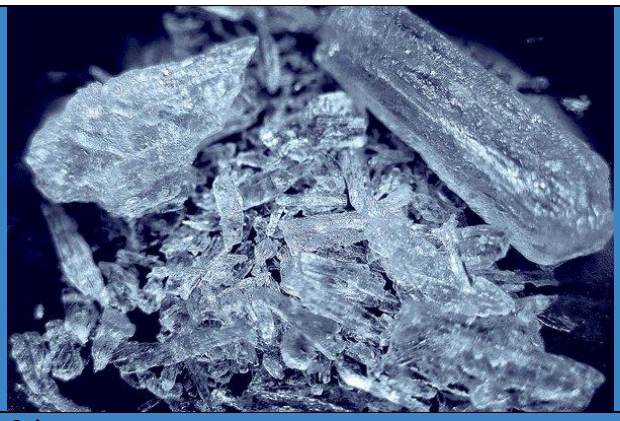
File Name	145382519_meth-rgb_t600[1].jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/145382519_meth-rgb_t600[1].jpg
File Size	88579
Created Time	2019-01-31 09:10:20 SGT
Accessed Time	2019-01-31 09:10:20 SGT
Modified Time	2019-01-31 09:10:21 SGT
MD5 Hash	f3905b24b628f5eb076c8524bcf

File Name	635744944987080757-Escobar-Meth-pic-1[1].jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/635744944987080757-Escobar-Meth-pic-1[1].jpg
File Size	46759
Created Time	2019-01-31 09:10:20 SGT
Accessed Time	2019-01-31 09:10:20 SGT
Modified Time	2019-01-31 09:10:20 SGT
MD5 Hash	4de09ccb91007d47a8a766a34a74679d



File Name	7b74bb6d-2118-405a-9c51-8a7a5925c5e7-meth-12_1[1].jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/$OrphanFiles/7b74bb6d-2118-405a-9c51-8a7a5925c5e7-meth-12_1[1].jpg
File Size	47109
Created Time	2019-01-31 09:12:34 SGT
Accessed Time	2019-01-31 09:12:34 SGT
Modified Time	2019-01-31 09:12:34 SGT
MD5 Hash	e1564ce3e8755da311e52be6af3e2bbe

File Name	crys1.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys1.jpg
File Size	39684
Created Time	2019-01-31 09:10:09 SGT
Accessed Time	2019-01-31 09:10:09 SGT
Modified Time	2019-01-31 09:10:09 SGT
MD5 Hash	5c580e37403cc6138497a23e9b0b9464



File Name	crys2.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys2.jpg
File Size	88579
Created Time	2019-01-31 09:10:31 SGT
Accessed Time	2019-01-31 09:10:32 SGT
Modified Time	2019-01-31 09:10:32 SGT
MD5 Hash	9d005f3905b24b628f5eb076c8524bcf

File Name	crys3.jpg
Analysis	
File Path	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys3.jpg
File Size	7330
Created Time	2019-01-31 09:12:41 SGT
Accessed Time	2019-01-31 09:12:41 SGT
Modified Time	2019-01-31 09:12:41 SGT
MD5 Hash	4eb61426baf81385864053359981609 6

Google Drive Link: https://drive.google.com/drive/folders/1GDlg46Ua9FXDF1uRC_7R-nk_EVSSy4lr?usp=sharing